Find a Fourier Series for the function defined by the equation

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ফ্যুরিয়ার ধারা

সমাধান

$f(x) = x, -\pi < x < \pi$

1(i) দেওয়া আছে (Given that)

$f(x) = x, -\pi < x < \pi.$

$f(-x) = -x = -f(x).$

$\Rightarrow f(x)$ বিজোড় ফাংশন, কাজেই ইহার ফ্যুরিয়ার ধারা হইল [$\Rightarrow f($ an odd function, so its Fourier series is]

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx, \ldots (1)$$

যখন $b_n = \dfrac{2}{\pi}\displaystyle\int_0^{\pi} f(x)\sin nx\, dx,$

বা $b_n = \dfrac{2}{\pi}\displaystyle\int_0^{\pi} x \sin nx\, dx,$

$$= \frac{2}{\pi}\left[x \cdot \frac{(-\cos nx)}{n} - (1)\frac{(-\sin nx)}{n^2}\right]_0^{\pi}.$$

$$= \frac{2}{\pi}\left[-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^{\pi}.$$

$$= \frac{2}{\pi}\left[\left(\frac{-\pi\cos n\pi}{n} + 0\right) - 0\right];$$ যেহেতু $\sin n\pi = 0, \sin 0 = ($

$$= -\frac{2\cos n\pi}{n} = \frac{-2(-1)^n}{n},$$

এখন b_n এর মান (1) নং এ বসাইয়া পাই, [Now putting the valu of b_n in (1) then we get]

$$f(x) = -\sum_{n=1}^{\infty}\frac{2(-1)^n}{n}\sin nx,$$

$$= -2\sum_{n=1}^{\infty}\frac{(-1)^n \sin nx}{n}.$$

even $f(x) = f(-x)$

odd $f(-x) = -f(x)$

$$= -2\left(-\frac{\sin x}{1} + \frac{\sin 2x}{2} - \frac{\sin 3x}{3} + \ldots\right).$$

$$= 2\left(\frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \ldots\right).$$

(ii). দেওয়া আছে (Given that)

$f(x) = |x|, -\pi < x < \pi.$

$\therefore f(-x) = |-x| = |x| = f(x)$

$\Rightarrow$ f(x) জোড় ফাংশন, কাজেই ইহার ফ্যুরিয়ার ধারা হইল [$\Rightarrow$ f(x) is an even function, so its Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx, \ldots (1)$$

যখন $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx$ এবং $a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx,$

$$\therefore a_0 = \frac{2}{\pi}\int_0^{\pi} |x|\,dx,$$

$$= \frac{2}{\pi}\int_0^{\pi} x\,dx, \text{ যেহেতু } x \geq 0.$$

$$= \frac{2}{\pi}\left[\frac{x^2}{2}\right]_0^{\pi} = \frac{2}{\pi}\left(\frac{\pi^2}{2} - 0\right) = \pi.$$

এবং $a_n = \frac{2}{\pi}\int_0^{\pi} |x| \cos nx\,dx,$

$$= \frac{2}{\pi}\int_0^{\pi} x\cos nx\,dx, \text{ যেহেতু } x \geq 0.$$

$$= \frac{2}{\pi}\left[x\,\frac{\sin nx}{n} - (1)\,\frac{-\cos nx}{n^2}\right]_0^{\pi}.$$

$$= \frac{2}{\pi}\left[\frac{x\sin nx}{n} + \frac{\cos nx}{n^2}\right]_0^{\pi}.$$

$$= \frac{2}{\pi}\left\{\left(0 + \frac{\cos n\pi}{n^2}\right) - \left(0 + \frac{\cos 0}{n^2}\right)\right\}.$$

$$= \frac{2}{\pi n^2}\{\cos n\pi - 1\},$$

$$= \frac{2\{(-1)^n - 1\}}{\pi n^2}.$$

$$\therefore (1) \Rightarrow f(x) = \frac{\pi}{2} + \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{\{(-1)^n - 1\}\cos nx}{n^2}.$$

বা $f(x) = \frac{\pi}{2} + \frac{2}{\pi}\left(\frac{-2\cos x}{1^2} + 0 + \frac{-2\cos 3x}{3^2} + 0 + \frac{-2\cos 5x}{5^2} + 0 + \ldots\right).$

বা $f(x) = \frac{\pi}{2} - \frac{4}{\pi}\left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \ldots\right).$

(iii). দেওয়া আছে (Given that) $f(x) = x^2, -\pi < x < \pi$ [illegible]

$f(x) = x^2, -\pi < x < \pi.$

$\therefore f(-x) = (-x)^2 = x^2 = f(x).$

$\Rightarrow$ f(x) যুগ্ম ফাংশন, কাজেই ইহার ফ্যুরিয়ার ধারা হইল [$\Rightarrow$ f(x) is an even function, so its Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx, \ldots (1).$$

যখন $a_0 = \frac{2}{\pi}\int_0^{\pi} f(x)\,dx$ এবং $a_n = \frac{2}{\pi}\int_0^{\pi} f(x)\cos nx\,dx.$

$$\therefore a_0 = \frac{2}{\pi}\int_0^{\pi} x^2\,dx = \frac{2}{\pi}\left[\frac{x^3}{3}\right]_0^{\pi} = \frac{2}{\pi}\left(\frac{\pi^3}{3} - 0\right).$$

$$= \frac{2\pi^2}{3}.$$

এবং $a_n = \frac{2}{\pi}\int_0^{\pi} x^2 \cos nx\,dx,$

বা $a_n = \frac{2}{\pi}\left[x^2 \cdot \frac{\sin nx}{n} - (2x)\,\frac{(-\cos nx)}{n^2} + (2)\cdot\frac{(-\sin nx)}{n^3}\right]_0^{\pi}.$

$$=\frac{2}{\pi}\left[\frac{x^2 \sin nx}{n}+\frac{2x\cos nx}{n^2}-\frac{2\sin nx}{n^3}\right]_0^{\pi}.$$

$$=\frac{2}{\pi}\left\{\left(0+\frac{2\pi\cos n\pi}{n^2}-0\right)-0\right\}, \text{ যেহেতু } \sin n\pi=0.$$

$$=\frac{4\cos n\pi}{n^2}=\frac{4(-1)^n}{n^2}.$$

এখন f(x), a_0 এবং a_n এর মান (1) নং এ বসাইয়া পাই, [Now putting the values of f(x), a_0 and a_n in (1) then we get]

$$x^2=\frac{1}{2}\cdot\frac{2\pi^2}{3}+4\sum_{n=1}^{\infty}\frac{(-1)^n\cos nx}{n^2}.$$

$$\text{বা } x^2=\frac{\pi^2}{3}+4\left(\frac{-\cos x}{1^2}+\frac{\cos 2x}{2^2}-\frac{\cos 3x}{3^2}+\dots\right).$$

$$\therefore x^2=\frac{\pi^2}{3}-4\left(\frac{\cos x}{1^2}-\frac{\cos 2x}{2^2}+\frac{\cos 3x}{3^2}-\dots\right)$$

2(ii). দেওয়া আছে (Given that)

$$f(x)=\begin{cases}-1, \text{ যখন } -\pi<x<0\\ 1, \text{ যখন } 0<x<\pi\end{cases} \quad \dots (1).$$

$$\pi/4=1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\dots$$

আমরা জানি $-\pi<x<\pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল [We know, in the interval $-\pi<x<x$, the Fourier series is]

$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n\cos nx+b_n\sin nx), \dots (2)$$

$$\text{এখন } a_0=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\,dx,$$

$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0}f(x)\,dx+\int_{0}^{\pi}f(x)\,dx\right\}.$$

$$=\frac{1}{\pi}\left\{-\int_{-\pi}^{0}1\,dx+\int_{0}^{\pi}1\,dx\right\}, \text{ (1) নং দ্বারা ।}$$

Find a Fourier series for the function defined by the equation $f(x)=\begin{cases}-1 & \text{for } -\pi<x<0\\ 1 & \text{for } 0<x<\pi\end{cases}$ Hence prove that $\pi/4=1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}+\dots$

$$=\frac{1}{\pi}\left[-[x]_{-\pi}^{0}+[x]_{0}^{\pi}\right].$$

$$=\frac{1}{\pi}\{-(0+\pi)+(\pi-0)\}.$$

$$=0.$$

$$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\cos nx\,dx.$$

$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0}f(x)\cos nx\,dx+\int_{0}^{\pi}f(x)\cos nx\,dx\right\}.$$

$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0}-1\cos nx\,dx+\int_{0}^{\pi}1.\cos nx\,dx\right\}, \text{ (1) নং হইতে}$$

$$=\frac{1}{\pi}\left\{-\left[\frac{\sin nx}{n}\right]_{-\pi}^{0}+\left[\frac{\sin nx}{n}\right]_{0}^{\pi}\right\}.$$

$$=\frac{1}{\pi}\left\{-\frac{1}{n}(0)+\frac{1}{n}(0)\right\}, \text{ যেহেতু } \sin n\pi=0.$$

$$=0.$$

$$\text{এবং } b_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\sin nx\,dx.$$

$$\text{বা } b_n=\frac{1}{\pi}\left\{\int_{-\pi}^{0}f(x)\sin nx\,dx+\int_{0}^{\pi}f(x)\sin nx\,dx\right\}.$$

$$=\frac{1}{\pi}\left\{-\int_{-\pi}^{0}1.\sin nx\,dx+\int_{0}^{\pi}1\sin nx\,dx\right\}.$$

$$=\frac{1}{\pi}\left\{\left[\frac{\cos nx}{n}\right]_{-\pi}^{0}-\left[\frac{\cos nx}{n}\right]_{0}^{\pi}\right\}.$$

$$=\frac{1}{\pi}\left\{\frac{1}{n}(1-\cos n\pi)-\frac{1}{n}(\cos n\pi-1)\right\}; \text{ যেহেতু } \boxed{\cos(-\theta)=\cos\theta}.$$

$$=\frac{1}{n\pi}(\cancel{1}-\cos n\pi-\cos n\pi+\cancel{1})=\frac{2}{n\pi}(1-\cos n\pi).$$

$$=\frac{2}{n\pi}\{1-(-1)^n\}.$$

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$$(2) \Rightarrow f(x) = 0 + \sum_{n=1}^{\infty} \frac{2\{1-(-1)^n\} \sin nx}{n\pi}.$$

$$\text{বা } f(x) = \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{\{1-(-1)^n\} \sin nx}{n}.$$

$$\text{বা } f(x) = \frac{2}{\pi} \left(\frac{2 \sin x}{1} + 0 + \frac{2 \sin 3x}{3} + 0 + \frac{2 \sin 5x}{5} + \ldots\right).$$

$$\therefore f(x) = \frac{4}{\pi} \left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \ldots\right) , \ldots (3)$$

দ্বিতীয় অংশ :

এখন (3) নং এ $x = \pi/2$ বসাইয়া পাই [Now in (3), we put $x = \pi/2$ then we get]

$$f\left(\frac{\pi}{2}\right) = \frac{4}{\pi} \left(\frac{\sin \pi/2}{1} + \frac{\sin 3\pi/2}{3} + \frac{\sin 5\pi/2}{5} + \ldots\right).$$

$$\text{বা } 1 = \frac{4}{\pi} \left(\frac{1}{1} - \frac{1}{3} + \frac{1}{5} - \ldots\right);$$ যেহেতু $f(x) = 1,\ 0 < x < \pi$

$$\therefore \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \ldots$$ প্রমাণিত।

$\pi/4 = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7}$

(ii) দেওয়া আছে (Given that)

$$f(x) = \begin{cases} -k, & \text{যখন } -\pi < x < 0 \\ k, & \text{যখন } 0 < x < \pi \end{cases} \ldots (1)$$

আমরা জানি, $-\pi < x < \pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [We know that in the interval $-\pi < x < \pi$, the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx), \ldots (2)$$

$$\text{যখন } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x)\, dx.$$

$$= \frac{1}{\pi} \left\{\int_{-\pi}^{0} f(x)\, dx + \int_{0}^{\pi} f(x)\, dx\right\}.$$

$$a_0 = \frac{1}{\pi} \left\{-\int_{-\pi}^{0} k\, dx + \int_{0}^{\pi} k\, dx\right\}.$$

$$= \frac{1}{\pi} \left\{-k\, [x]_{-\pi}^{0} + k\, [x]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi} \{-k\,(0 + \pi) + k\,(\pi - 0)\} = 0.$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx\, dx.$$

$$= \frac{1}{\pi} \left\{\int_{-\pi}^{0} f(x) \cos nx\, dx + \int_{0}^{\pi} f(x) \cos nx\, dx\right\}.$$

$$= \frac{1}{\pi} \left\{-\int_{-\pi}^{0} k \cos nx\, dx + \int_{0}^{\pi} k \cos nx\, dx\right\}.$$

$$= \frac{1}{\pi} \left\{-k \left[\frac{\sin nx}{n}\right]_{-\pi}^{0} + k \left[\frac{\sin nx}{n}\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi} \left\{-\frac{k}{n}(0) + \frac{k}{n}(0)\right\},$$ যেহেতু $\sin(-n\pi) = -\sin n\pi = 0$, $\sin 0 = 0$.

$$= 0.$$

$$\text{এবং } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx\, dx.$$

$$= \frac{1}{\pi} \left\{\int_{-\pi}^{0} f(x) \sin nx\, dx + \int_{0}^{\pi} f(x) \sin nx\, dx\right\}.$$

$$= \frac{1}{\pi} \left\{-\int_{-\pi}^{0} k \sin nx\, dx + \int_{0}^{\pi} k \sin nx\, dx\right\}.$$

$$= \frac{1}{\pi} \left\{k \left[\frac{\cos nx}{n}\right]_{-\pi}^{0} - k \left[\frac{\cos nx}{n}\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi} \left\{\frac{k}{n}(1 - \cos n\pi) - \frac{k}{n}(\cos n\pi - 1)\right\}.$$

$$= \frac{k}{n\pi}(1 - \cos n\pi - \cos n\pi + 1).$$

$$= \frac{2k}{n\pi}(1 - \cos n\pi) = \frac{2k\{1-(-1)^n\}}{n\pi}$$

$$\Rightarrow f(x) = 0 + \sum_{n=1}^{\infty} \frac{2k}{n\pi}\{1-(-1)^n\} \sin nx.$$

$$\text{বা } f(x) = \frac{2k}{\pi}\sum_{n=1}^{\infty} \frac{\{1-(1)^n\} \sin nx}{n}.$$

$$\text{বা } f(x) = \frac{2k}{\pi}\left(\frac{2\sin x}{1} + 0 + \frac{2\sin 3x}{3} + 0 + \frac{2\sin 5x}{5} + \ldots\right).$$

$$\text{বা } f(x) = \frac{4k}{\pi}\left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \ldots\right) \ldots (3).$$

দ্বিতীয় অংশ ঃ এখন (3) নং $x = \pi/2$ বসাইয়া পাই [Now we put $x = \pi/2$ in (3) then we get]

$$f\left(\frac{\pi}{2}\right) = \frac{4k}{\pi}\left(\frac{\sin \pi/2}{1} + \frac{\sin 3\pi/2}{3} + \frac{\sin 5\pi/2}{5} + \ldots\right).$$

বা $k = \frac{4k}{\pi}\left(1 - \frac{1}{3} + \frac{1}{5} - \ldots\right)$; যেহেতু $f(x) = k,\ 0 < x < \pi$.

$\therefore \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \ldots$ প্রমাণিত।

(iii). দেওয়া আছে (Given that)

$$f(x) = \begin{cases} -\pi/4, & \text{যখন } -\pi < x < 0 \\ \pi/4, & \text{যখন } 0 < x < \pi \end{cases}$$

আমরা জানি, $-\pi < x < \pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [We know, in the interval $-\pi < x < \pi$, the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx), \ldots (1)$$

যখন $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\, dx,$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\, dx + \int_{0}^{\pi} f(x)\, dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\int_{-\pi}^{0} \frac{\pi}{4}\, dx + \int_{0}^{\pi} \frac{\pi}{4}\, dx\right\}.$$

$$\frac{\pi}{4}\left[x\right]_{-\pi}^{0} + \frac{\pi}{4}\left[x\right]_{0}^{\pi}\Big\}.$$

$$= \frac{1}{\pi}\left\{-\frac{\pi}{4}(0+\pi) + \frac{\pi}{4}(\pi - 0)\right\}$$

$$= 0.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \cos nx\, dx,$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x) \cos nx\, dx + \int_{0}^{\pi} f(x) \cos nx\, dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\int_{-\pi}^{0} \frac{\pi}{4} \cos nx\, dx + \int_{0}^{\pi} \frac{\pi}{4} \cos nx\, dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\frac{\pi}{4}\left[\frac{\sin nx}{n}\right]_{-\pi}^{0} + \frac{\pi}{4}\left[\frac{\sin nx}{n}\right]_{0}^{\pi}\right\}.$$

$= \frac{1}{\pi}\left\{-\frac{\pi}{4n}(0) + \frac{\pi}{4n}(0)\right\}$. যেহেতু $\sin(-n\pi) = \sin n\pi = 0$.

$$= 0.$$

এবং $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx\, dx,$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x) \sin nx\, dx + \int_{0}^{\pi} f(x) \sin nx\, dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\int_{-\pi}^{0} \frac{\pi}{4} \sin nx\, dx + \int_{0}^{\pi} \frac{\pi}{4} \sin nx\, dx\right\}.$$

$$= \frac{1}{\pi}\left\{\frac{\pi}{4}\left[\frac{\cos nx}{n}\right]_{-\pi}^{0} + \frac{\pi}{4}\left[\frac{-\cos nx}{n}\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{\frac{\pi}{4n}(1 - \cos n\pi) + \frac{\pi}{4n}(-\cos n\pi + 1)\right\}.$$

$$= \frac{1}{4n}(1 - \cos n\pi - \cos n\pi + 1).$$

$$= \frac{2}{4n}(1 - \cos n\pi) = \frac{1-(-1)^n}{2n}.$$

$$\therefore (2) \Rightarrow f(x) = 0 + \frac{1}{2}\sum_{n=0}^{\infty} \frac{\{1-(-1)^n\}\sin nx}{n}.$$

$$\text{বা } f(x) = \frac{1}{2}\left(\frac{2\sin x}{1} + 0 + \frac{2\sin 3x}{3} + 0 + \frac{2\sin 5x}{5} + \ldots\right).$$

$$\text{বা } f(x) = \left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \ldots\right).$$

বিঃ দ্রঃ বইয়ের উত্তর মালায় –4 হইবে না।

(iv). দেওয়া আছে (Given that)

$$f(x) = \begin{cases} -1, \text{ যখন } -\pi < x < 0 \\ 0, \text{ যখন } \quad x = 0 \\ 1, \text{ যখন } -0 < x < \pi \end{cases} \quad \ldots (1)$$

আমরা জানি, $-\pi < x < \pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল [we know, in the interval $-\pi < x < \pi$, the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx), \ldots (2)$$

$$\text{যখন } a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx,$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\,dx + \int_{0}^{\pi} f(x)\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\int_{-\pi}^{0} 1\,dx + \int_{0}^{\pi} 1\,dx\right\}, \text{ (1) নং দ্বারা,}$$

$$= \frac{1}{\pi}\left\{-\left[x\right]_{-\pi}^{0} + \left[x\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\{-(0+\pi) + (\pi - 0)\},$$

$$= 0.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx,$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\cos nx\,dx + \int_{0}^{\pi} f(x)\cos nx\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\int_{-\pi}^{0} 1.\cos nx\,dx + \int_{0}^{\pi} 1\cos nx\,dx\right\}, \text{ (1) নং দ্বারা।}$$

$$= \frac{1}{\pi}\left\{-\left[\frac{\sin nx}{n}\right]_{-\pi}^{0} + \left[\frac{\sin nx}{n}\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{-\frac{1}{n}(0) + \frac{1}{n}(0)\right\}, \text{ যেহেতু } \sin(-n\pi) = -\sin n\pi = 0.$$

$$= 0.$$

$$\text{এবং } b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx,$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\sin nx\,dx + \int_{0}^{\pi} f(x)\sin nx\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\int_{-\pi}^{0} 1\sin nx\,dx + \int_{0}^{\pi} 1\sin nx\,dx\right\}, \text{ (1) নং দ্বারা।}$$

$$= \frac{1}{\pi}\left\{\left[\frac{\cos nx}{n}\right]_{-\pi}^{0} - \left[\frac{\cos nx}{n}\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{\frac{1}{n}(1-\cos n\pi) - \frac{1}{n}(\cos n\pi - 1)\right\}.$$

$$= \frac{1}{\pi n}(1 - \cos n\pi - \cos n\pi + 1) = \frac{2}{n\pi}(1-\cos n\pi).$$

$$= \frac{2\{1-(-1)^n\}}{n\pi}.$$

$$\therefore (2) \Rightarrow f(x) = 0 + \sum_{n=1}^{\infty} \frac{2\{1-(-1)^n\}\sin nx}{\pi n},$$

$$\text{বা } f(x) = \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{\{1-(-1)^n\}\sin nx}{n}.$$

$$\text{বা } f(x) = \frac{2}{\pi}\left(\frac{2\sin x}{1} + 0 + \frac{2\sin 3x}{3} + 0 + \frac{2\sin 5x}{5} + \ldots\right).$$

$$\therefore\ f(x) = \frac{4}{\pi}\left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \dots\right).$$

(v). দেওয়া আছে (Given that)

$$f(x) = \begin{cases} -1, & -\pi < x < -\pi/2 \\ 0, & -\pi/2 < x < \pi/2 \\ 1, & \pi/2 < x < \pi \end{cases} \quad \dots (1)$$

আমরা জানি, $-\pi < x < \pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [We know, in the interval $-\pi < x < \pi$, the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx), \dots (2)$$

যখন $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$,

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{-\pi/2} f(x)\,dx + \int_{-\pi/2}^{\pi/2} f(x)\,dx + \int_{\pi/2}^{\pi} f(x)\,dx\right\}.$$

$$= \frac{1}{\pi} = \left\{-\int_{-\pi}^{-\pi/2} 1\,dx + \int_{-\pi/2}^{\pi/2} 0\,dx + \int_{\pi/2}^{\pi} 1\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\left[x\right]_{-\pi}^{-\pi/2} + 0 + \left[x\right]_{\pi/2}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{-\left(-\frac{\pi}{2} + \pi\right) + \left(\pi - \frac{\pi}{2}\right)\right\} = \frac{1}{\pi}\left(\frac{\pi}{2} - \pi + \pi - \frac{\pi}{2}\right).$$

$$= 0.$$

$$a_n = \frac{1}{n}\int_{-\pi}^{\pi} f(x)\cos nx\,dx,$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{-\pi/2} f(x)\cos nx\,dx + \int_{-\pi/2}^{\pi/2} f(x)\cos nx\,dx + \int_{\pi/2}^{\pi} f(x)\cos nx\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\int_{-\pi}^{-\pi/2} 1.\cos nx\,dx + \int_{-\pi/2}^{\pi/2} 0.\cos nx\,dx + \int_{\pi/2}^{\pi} 1.\cos nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{-\left[\frac{\sin nx}{n}\right]_{-\pi}^{-\pi/2} + 0 + \left[\frac{\sin nx}{n}\right]_{\pi/2}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{-\frac{1}{n}(-\sin n\pi/2 + \sin n\pi) + \frac{1}{n}(\sin n\pi - \sin n\pi/2)\right\}.$$

$$= \frac{1}{n\pi}(\sin n\pi/2 + 0 + 0 - \sin n\pi/2) = 0.$$

এবং $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$.

বা $b_n = \frac{1}{\pi}\left\{\int_{-\pi}^{-\pi/2} f(x)\sin nx\,dx + \int_{-\pi/2}^{\pi/2} f(x)\sin nx\,dx + \int_{\pi/2}^{\pi} f(x)\sin nx\,dx\right\}$.

$$= \frac{1}{\pi}\left\{-\int_{-\pi}^{-\pi/2} 1.\sin nx\,dx + \int_{-\pi/2}^{\pi/2} 0.\sin nx\,dx + \int_{\pi/2}^{\pi} 1.\sin nx\,dx\right\}$$

$$= \frac{1}{\pi}\left\{\left[\frac{\cos nx}{n}\right]_{-\pi}^{-\pi/2} + 0 - \left[\frac{\cos nx}{n}\right]_{\pi/2}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{\frac{1}{n}\left(\cos\frac{n\pi}{2} - \cos n\pi\right) - \frac{1}{n}\left(\cos n\pi - \cos\frac{n\pi}{2}\right)\right\}.$$

$$= \frac{1}{\pi n}\left(\cos\frac{n\pi}{2} - \cos n\pi - \cos n\pi + \cos\frac{n\pi}{2}\right).$$

$$= \frac{2}{\pi n}\left(\frac{\cos n\pi}{2} - \cos n\pi\right).$$

$$= \frac{2}{\pi n}\left\{\frac{\cos n\pi}{2} - (-1)^n\right\}.$$

$$\therefore (2) \Rightarrow f(x) = 0 + \frac{2}{\pi}\sum_{n=1}^{\infty} \frac{\{\cos n\pi/2 - (-1)^n\}\sin nx}{n},$$

$$\text{বা } f(x) = \frac{2}{\pi}\Big\{\frac{(0+1)\sin x}{1} + \frac{(-1-1)\sin 2x}{2} + \frac{(0+1)\sin 3x}{3} + \frac{(1-1)\sin 4x}{4} + \frac{(0+1)\sin 5x}{5} + \frac{(-1-1)\sin 6x}{6} + \frac{(0+1)\sin 7x}{7} + \frac{(1-1)\sin 8x}{8} + \frac{(0+1)\sin 9x}{9} + \frac{(-1-1)\sin 10x}{10} + \dots\Big\}.$$

$$\text{বা } f(x) = \frac{2}{\pi}\left(\frac{\sin x}{1} - \frac{2\sin 2x}{2} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} - \frac{2\sin 6x}{6} + \frac{\sin 7x}{7} + \frac{\sin 9x}{9} - \frac{2\sin 10x}{10} + \dots\right).$$

$$\text{বা } f(x) = \frac{2}{\pi}\left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \frac{\sin 7x}{7} + \dots\right) - \frac{4}{\pi}\left(\frac{\sin 2x}{2} + \frac{\sin 6x}{6} + \frac{\sin 10x}{10} + \dots\right) \quad \dots (3)$$

২য় অংশ : $f(x)$ ফাংশনটি $x = \frac{\pi}{2}$ বিন্দুতে বিচ্ছিন্ন [The function $f(x)$ is discontinuous at the point $x = \pi/2$].

এখানে $f(\pi/2 - 0) = 0$ এবং $f(\pi/2 + 0) = 1$.

$$\therefore f(\pi/2) = \frac{1}{2}\{f(\pi/2 - 0) + f(\pi/2 + 0)\}$$

$$= \frac{1}{2}(0+1) = \frac{1}{2}, \dots (4)$$

এখন (3) নং এ $x = \pi/2$ বসাইয়া পাই, [Now in (3), we put $x = \pi/2$ then we get]

$$f(\pi/2) = \frac{2}{\pi}\left(\frac{\sin \pi/2}{1} + \frac{\sin 3\pi/2}{3} + \frac{\sin 5\pi/2}{5} + \frac{\sin 7\pi/2}{7} + \dots\right) - \frac{4}{\pi}\left(\frac{\sin\pi}{2} + \frac{\sin 3\pi}{6} + \frac{\sin 5\pi}{10} + \dots\right).$$

$$\text{বা } \frac{1}{2} = \frac{2}{\pi}\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots\right) - \frac{4}{\pi}(0); \text{ (4) নং দ্বারা।}$$

$$\therefore \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots .$$

$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

3(i). দেওয়া আছে (Given that)

$$f(x) = \begin{cases} -\pi, \text{ যখন } -\pi < x < 0 \\ x, \text{ যখন } \quad 0 < x < \pi \end{cases} \quad \dots (1)$$

আমরা জানি $-\pi < x < \pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [we know, in the interval $-\pi < x < \pi$, the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx), \dots (2)$$

$$\text{যখন } a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx,$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\,dx + \int_0^{\pi} f(x)\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\int_{-\pi}^{0}\pi\,dx + \int_0^{\pi} x\,dx\right\}. \text{ (1) নং দ্বারা।}$$

$$= \frac{1}{\pi}\left\{-\pi\,[x]_{-\pi}^{0} + \left[\frac{x^2}{2}\right]_0^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{-\pi(0+\pi) + \frac{\pi^2}{2}\right\} = \frac{1}{\pi}\left(-\pi^2 + \frac{\pi^2}{2}\right)$$

$$= \frac{1}{\pi}\left(-\frac{\pi^2}{2}\right) = -\frac{\pi}{2}.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx,$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\cos nx\,dx + \int_0^{\pi} f(x)\cos nx\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\int_{-\pi}^{0}\pi\cos nx\,dx + \int_0^{\pi} x\cos nx\,dx\right\}; \text{ (1) নং দ্বারা।}$$

$$= \frac{1}{\pi}\left\{-\pi\left[\frac{\sin nx}{n}\right]_0^{\pi} + \left[x\left(\frac{\sin nx}{n}\right) - (1).\left(\frac{-\cos nx}{n^2}\right)\right]_0^{\pi}\right\}$$

$$=\frac{1}{\pi}\left\{\frac{-\pi}{n}(0)+\left[\frac{x\sin nx}{n}+\frac{\cos nx}{n^2}\right]_0^\pi\right\}.$$

$$=\frac{1}{\pi}\left\{0+\left(0+\frac{\cos n\pi}{n^2}\right)-\left(0+\frac{1}{n^2}\right)\right\}.$$

$$=\frac{1}{\pi n^2}(\cos n\pi-1)=\frac{(-1)^n-1}{\pi n^2}.$$

এবং $b_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx$,

$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\sin nx\,dx+\int_0^\pi f(x)\sin nx\,dx\right\}.$$

$$=\frac{1}{\pi}\left\{-\int_{-\pi}^{0}\pi\sin nx\,dx+\int_0^\pi x\sin nx\,dx\right\}, \text{ (1) নং দ্বারা।}$$

$$=\frac{1}{\pi}\left\{\pi\left[\frac{\cos nx}{n}\right]_{-\pi}^{0}+\left[x\frac{(-\cos nx)}{n}-(1)\frac{(-\sin nx)}{n^2}\right]_0^\pi\right\}.$$

$$=\frac{1}{\pi}\left\{\frac{\pi}{n}(1-\cos n\pi)+\left[\frac{-x\cos nx}{n}+\frac{\sin nx}{n^2}\right]_0^\pi\right\}.$$

$$=\frac{1}{\pi}\left\{\frac{\pi}{n}(1-\cos n\pi)-\frac{\pi\cos n\pi}{n}+0\right\}.$$

$$=\frac{\pi}{\pi n}(1-2\cos n\pi)=\frac{1-2(-1)^n}{n}.$$

$$\therefore (2)\Rightarrow f(x)=\frac{-\pi}{4}+\sum_{n=1}^{\infty}\frac{\{(-1)^n-1\}\cos nx}{\pi n^2}+\sum_{n=1}^{\infty}\frac{\{1-2(-1)^n\}\sin nx}{n}.$$

$$\text{বা } f(x)=\frac{-\pi}{4}+\frac{1}{\pi}\left(\frac{-2\cos x}{1^2}+0+\frac{-2\cos 3x}{3^2}+0+\frac{-2\cos 5x}{5^2}+\dots\right)$$
$$+\left(\frac{3\sin x}{1}-\frac{\sin 2x}{2}+\frac{3\sin 3x}{3}-\frac{\sin 4x}{4}+\dots\right)$$

$$\text{বা } f(x)=\frac{-\pi}{4}-\frac{2}{\pi}\left(\frac{\cos x}{1^2}+\frac{\cos 3x}{3^2}+\frac{\cos 5x}{5^2}+\dots\right)$$
$$+\left(\frac{3\sin x}{1}-\frac{\sin 2x}{2}+\frac{3\sin 3x}{3}-\frac{\sin 4x}{4}+\dots\right)\quad \dots (3)$$

দ্বিতীয় অংশ : $f(x)$ ফাংশনটি $x = 0$ বিন্দুতে বিচ্ছিন্ন [The function $f(x)$ is discontinuous at the point $x = 0$].

এখানে $f(0-0) = -\pi$ এবং $f(0+0) = 0$.

$$\therefore f(0)=\frac{1}{2}\{f(0-0)+f(0+0)\}$$
$$=\frac{1}{2}(-\pi+0)$$
$$=-\frac{\pi}{2}, \dots (4).$$

এখন (3) নং এ $x = 0$ বসাইয়া পাই, [Now in (3), we put $x = 0$ then we get]

$$f(0)=-\frac{\pi}{4}-\frac{2}{\pi}\left(\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\dots\right)+0.$$

$$\text{বা } -\frac{\pi}{2}=-\frac{\pi}{4}-\frac{2}{\pi}\left(\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\dots\right). \text{ (4) নং দ্বারা।}$$

$$\text{বা } -\frac{\pi}{4}=-\frac{2}{\pi}\left(\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\dots\right).$$

$$\therefore \frac{\pi^2}{8}=\frac{1}{1^2}+\frac{1}{3^2}+\frac{1}{5^2}+\dots \text{ প্রমাণিত।}$$

(ii). দেওয়া আছে (Given that)

$$f(x)=\begin{cases}0, \text{ যখন } -\pi<x<0\\ x, \text{ যখন } \ 0<x<\pi\end{cases}\quad \dots (1)$$

আমরা জানি, $-\pi < x < \pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [we know, in the interval $-\pi < x < \pi$, the Fourier series is]

$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n\cos nx+b_n\sin nx). \dots (2)$$

যখন $a_0=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx$,

বা $a_0 = \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\,dx + \int_{0}^{\pi} f(x)\,dx\right\}$.

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\,dx + \int_{0}^{\pi} x\,dx\right\} = \frac{1}{\pi}\left\{0 + \left[\frac{x^2}{2}\right]_0^{\pi}\right\}.$$

$$= \frac{1}{\pi}\cdot\frac{\pi^2}{2} = \frac{\pi}{2}.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx,$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\cos nx\,dx + \int_{0}^{\pi} f(x)\cos nx\,dx\right\},$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\cos nx\,dx + \int_{0}^{\pi} x\cos nx\,dx\right\}, \text{ (1) নং দ্বারা।}$$

$$= \frac{1}{\pi}\left\{0 + \left[x\cdot\frac{\sin nx}{n} - (1)\frac{-\cos nx}{n^2}\right]_0^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left[\frac{x\sin nx}{n} + \frac{\cos nx}{n^2}\right]_0^{\pi}.$$

$$= \frac{1}{\pi}\left\{\left(0 + \frac{\cos n\pi}{n^2}\right) - \left(0 + \frac{1}{n^2}\right)\right\}.$$

$$= \frac{1}{\pi n^2}(\cos n\pi - 1) = \frac{(-1)^n - 1}{\pi n^2}.$$

এবং $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx,$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\sin nx\,dx + \int_{0}^{\pi} f(x)\sin nx\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\sin nx\,dx + \int_{0}^{\pi} x\sin nx\,dx\right\}, \text{ (1) নং দ্বারা।}$$

$$= \frac{1}{\pi}\left\{0 + \left[\frac{x(-\cos nx)}{n} - \frac{(1)(-\sin nx)}{n^2}\right]\right\}.$$

$$= \frac{1}{\pi}\left[\frac{-x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^{\pi}.$$

$$= \frac{1}{\pi}\left\{\left(\frac{-\pi\cos n\pi}{n} + 0\right) - (0)\right\} = \frac{-\cos n\pi}{n} = \frac{-(-1)^n}{n}.$$

$$\therefore (2) \Rightarrow f(x) = \frac{\pi}{4} + \sum_{n=1}^{\infty}\frac{\{(-1)^n - 1\}\cos nx}{\pi n^2} - \sum_{n=1}^{\infty}\frac{(-1)^n\sin nx}{n},$$

$$\text{বা } f(x) = \frac{\pi}{4} + \frac{1}{\pi}\left(\frac{-2\cos x}{1^2} + 0 - \frac{2\cos 3x}{3^2} + 0 - \frac{2\cos 5x}{5^2} + \dots\right)$$
$$- \left(\frac{-\sin x}{1} + \frac{\sin 2x}{2} - \frac{\sin 3x}{3} + \dots\right).$$

$$\text{বা } f(x) = \frac{\pi}{4} - \frac{2}{\pi}\left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots\right)$$
$$+ \left(\frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} \dots\right)$$

(iii) দেওয়া আছে (Given that)

$$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ \pi x/4, & 0 < x < \pi \end{cases} \quad \dots (1)$$

আমরা জানি, $-\pi < x < \pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [we know, in the interval $-\pi < x < \pi$, the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n\cos nx + b_n\sin nx) \dots (1)$$

যখন $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx,$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\,dx + \int_{0}^{\pi} f(x)\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\,dx + \int_{0}^{\pi}\frac{\pi x}{4}\,dx\right\}, \text{ (1) নং দ্বারা।}$$

$$= \frac{1}{\pi}\left\{0 + \frac{\pi}{4}\left[\frac{x^2}{2}\right]_0^{\pi}\right\} = \frac{1}{\pi}\cdot\frac{\pi^3}{8} = \frac{\pi^2}{8}.$$

$\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5}$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx.$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\cos nx\,dx + \int_{0}^{\pi} f(x)\cos nx\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\cos nx\,dx + \int_{0}^{\pi} \frac{\pi x}{4}\cos nx\,dx\right\};$$ (1) নং দ্বারা।

$$= \frac{1}{\pi}\left\{0 + \frac{\pi}{4}\left[\frac{x\sin nx}{n} - (1)\frac{(-\cos nx)}{n^2}\right]_0^{\pi}\right\}.$$

$$= \frac{1}{4}\left[\frac{x\sin nx}{n} + \frac{\cos nx}{n^2}\right]_0^{\pi} = \frac{1}{4}\left\{\left(0 + \frac{\cos n\pi}{n^2}\right) - \left(0 + \frac{1}{n^2}\right)\right\}.$$

$$= \frac{(-1)^n - 1}{4n^2}.$$

এবং $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx.$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\sin nx\,dx + \int_{0}^{\pi} f(x)\sin nx\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\sin nx\,dx + \int_{0}^{\pi} \frac{\pi x}{4}\sin nx\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{0 + \frac{\pi}{4}\left[\frac{x(-\cos nx)}{n} - \frac{(1)(-\sin nx)}{n^2}\right]_0^{\pi}\right\}.$$

$$= \frac{1}{4}\left[\frac{-x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^{\pi} = \frac{1}{4}\left\{\left(\frac{-\pi\cos n\pi}{n} + 0\right) - (0)\right\}.$$

$$= \frac{-\pi(-1)^n}{4n}.$$

$$\therefore (2) \Rightarrow f(x) = \frac{\pi^2}{16} + \sum_{n=1}^{\infty}\frac{\{(-1)^n - 1\}\cos nx}{4n^2} - \sum_{n=1}^{\infty}\frac{\pi(-1)^n \sin nx}{4n}.$$

$$\text{বা } f(x) = \frac{\pi^2}{16} + \frac{1}{4}\left(\frac{-2\cos x}{1^2} + 0 - \frac{2\cos 3x}{3^2} + 0 - \frac{2\cos 5x}{5^2} + \dots\right) - \frac{\pi}{4}\left(\frac{-\sin x}{1} + \frac{\sin 2x}{2} - \frac{\sin 3x}{3} + \dots\right).$$

$$\text{বা } f(x) = \frac{\pi^2}{16} - \frac{1}{2}\left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots\right) + \frac{\pi}{4}\left(\frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \dots\right) \dots (3)$$

দ্বিতীয় অংশ ঃ (a). $f(x)$ ফাংশনটি $x = \pm\pi$ বিন্দুতে বিচ্ছিন্ন. [The function $f(x)$ is discontinuous at the point $x = \pm\pi$].

এখানে $f(-\pi + 0) = 0$ এবং $f(\pi - 0) = \frac{\pi}{4}.\pi = \frac{\pi^2}{4}$.

$$\therefore f(\pi) = \frac{1}{2}\{f(-\pi + 0) + f(\pi - 0)\} = \frac{1}{2}\left(0 + \frac{\pi^2}{4}\right).$$

বা $f(\pi) = \frac{\pi^2}{8}$, ... (4)

এখন (3) নং এ $x = \pi$ বসাইয়া পাই [Now in (3) we put $x = \pi$ then we get]

$$f(\pi) = \frac{\pi^2}{16} - \frac{1}{2}\left(\frac{-1}{1^2} + \frac{-1}{3^2} + \frac{-1}{5^2} + \dots\right) + \frac{\pi}{4}(0).$$

$$\text{বা } \frac{\pi^2}{8} = \frac{\pi^2}{16} + \frac{1}{2}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots\right)$$ (4) নং দ্বারা।

$$\text{বা } \frac{\pi^2}{8} - \frac{\pi^2}{16} = \frac{1}{2}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots\right).$$

$$\therefore \frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

(b). আবার (3) নং এ $x = \pi/2$ বসাইয়া পাই [Again in (3) we put $x = \pi/2$ then we get]

$$f(\pi/2) = \frac{\pi^2}{16} - \frac{1}{2}(0) + \frac{\pi}{4}\left(\frac{1}{1} - 0 + \frac{-1}{3} + 0 + \frac{1}{5} + 0 + \frac{-1}{7} + \dots\right).$$

$$\text{বা } \frac{\pi}{4}\cdot\frac{\pi}{2} = \frac{\pi^2}{16} + \frac{\pi}{4}\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots\right);$$ যেহেতু $0 < x < \pi$ ব্যবধিতে $f(x) = \frac{\pi x}{4}$.

$$\text{বা } \frac{\pi^2}{8} - \frac{\pi^2}{16} = \frac{\pi}{4}\left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots\right).$$

$$\therefore \frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots.$$

(iv) দেওয়া আছে (Given that)-

$$f(x) = \begin{cases} -x, \text{ যখন } -\pi < x < 0 \\ x, \text{ যখন } \ 0 < x < \pi \end{cases} \quad \ldots (1)$$

আমরা জানি, $-\pi < x < \pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [We know, in the interval $-\pi < x < \pi$, the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx), \ldots (2)$$

যখন $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\, dx,$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\, dx + \int_{0}^{\pi} f(x)\, dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\int_{-\pi}^{0} x\, dx + \int_{0}^{\pi} x\, dx\right\}, \text{ (1) নং দ্বারা ।}$$

$$= \frac{1}{\pi}\left\{-\left[\frac{x^2}{2}\right]_{-\pi}^{0} + \left[\frac{x^2}{2}\right]_{0}^{\pi}\right\} = \frac{1}{\pi}\left\{-\frac{1}{2}(0-\pi^2) + \frac{1}{2}\left(\frac{\pi^2}{2}\right)\right\}.$$

$$= \frac{1}{\pi}\cdot\left(\frac{\pi^2}{2} + \frac{\pi^2}{2}\right) = \pi.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \cos nx\, dx,$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x) \cos nx\, dx + \int_{0}^{\pi} f(x) \cos nx\, dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\int_{-\pi}^{0} x \cos nx\, dx + \int_{0}^{\pi} x \cos nx\, dx\right\}, \text{ (1) নং দ্বারা ।}$$

$$= \frac{1}{\pi}\left\{-\left[x\frac{\sin nx}{n} - (1)\frac{(-\cos nx)}{n^2}\right]_{-\pi}^{0} + \left[x\frac{\sin nx}{n} - (1)\frac{(-\cos nx)}{n^2}\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{-\left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2}\right]_{-\pi}^{0} + \left[\frac{x \sin nx}{n} + \frac{\cos nx}{n^2}\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{-\left(0+\frac{1}{n^2}\right) + \left(0+\frac{\cos n\pi}{n^2}\right) + \left(0+\frac{\cos n\pi}{n^2}\right) - \left(0+\frac{1}{n^2}\right)\right\}.$$

$$= \frac{2}{\pi n^2}(\cos n\pi - 1) = \frac{2\{(-1)^n - 1\}}{\pi n^2}.$$

এবং $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \sin nx\, dx,$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x) \sin nx\, dx + \int_{0}^{\pi} f(x) \sin nx\, dx\right\}.$$

$$= \frac{1}{\pi}\left\{-\int_{-\pi}^{0} x \sin nx\, dx + \int_{0}^{\pi} x \sin nx\, dx\right\}, \text{ (1) নং দ্বারা ।}$$

$$= \frac{1}{\pi}\left\{-\left[\frac{x(-\cos nx)}{n} - \frac{(1)(-\sin nx)}{n^2}\right]_{-\pi}^{0} + \left[\frac{x(-\cos nx)}{n} - \frac{(1)(-\sin nx)}{n^2}\right]_{0}^{\pi}\right\}$$

$$= \frac{1}{\pi}\left\{-\left[\frac{-x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_{-\pi}^{0} + \left[\frac{-x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{-(0+0) + \left(\frac{\pi\cos n\pi}{n} + 0\right) + \left(\frac{-\pi\cos n\pi}{n} + 0\right) - (0)\right\}.$$

$$= \frac{1}{\pi}\left(\frac{\pi \cos n\pi}{n} - \frac{\pi \cos n\pi}{n}\right) = 0.$$

$$\therefore (2) \Rightarrow f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2\{(-1)^n - 1\} \cos nx}{\pi n^2} + \sum_{n=1}^{\infty} 0.\sin nx,$$

$$\text{বা } f(x) = \frac{\pi}{2} + \frac{2}{\pi}\left(\frac{-2\cos x}{1^2} + 0 + \frac{-2\cos 3x}{3^2} + 0 + \frac{-2\cos 5x}{5^2} + \ldots\right) + 0.$$

$$\text{বা } f(x) = \frac{\pi}{2} - \frac{4}{\pi}\left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \ldots\right).$$

4(i). দেওয়া আছে (Given that)

$$f(x) = \begin{cases} 2, & -2 < x < 0 \\ x, & 0 < x < 2 \end{cases} \quad \ldots (1)$$

আমরা জানি, $-2 < x < 2$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [We know, in he interval $-2 < x < 2$, the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left(a_n \cos\frac{n\pi x}{2} + b_n \sin\frac{n\pi x}{2}\right) \quad \ldots (2)$$

যখন $a_0 = \frac{1}{2}\int_{-2}^{2} f(x)\,dx.$

$$= \frac{1}{2}\left\{\int_{-2}^{0} f(x)\,dx + \int_{0}^{2} f(x)\,dx\right\}.$$

$$= \frac{1}{2}\left\{\int_{-2}^{0} 2\,dx + \int_{0}^{2} x\,dx\right\}.$$ (1) নং দ্বারা।

$$= \frac{1}{2}\left\{2\left[x\right]_{-2}^{0} + \left[\frac{x^2}{2}\right]_{0}^{2}\right\} = \frac{1}{2}\left\{2(0+2) + \frac{2^2}{2}\right\}.$$

$$= \frac{1}{2}(4+2) = 3.$$

$$a_n = \frac{1}{2}\int_{-2}^{2} f(x)\cos\frac{n\pi x}{2}\,dx.$$

$$= \frac{1}{2}\left\{\int_{-2}^{0} f(x)\cos\frac{n\pi x}{2}\,dx + \int_{0}^{2} f(x)\cos\frac{n\pi x}{2}\,dx\right\}.$$

$$= \frac{1}{2}\left\{\int_{-2}^{0} 2\cos\frac{n\pi x}{2}\,dx + \int_{0}^{2} x\cos\frac{n\pi x}{2}\,dx\right\}.$$ (1) নং দ্বারা।

$$= \frac{1}{2}\left\{2\left[\frac{\sin n\pi x/2}{n\pi/2}\right]_{-2}^{0} + \left[\frac{x.\sin n\pi x/2}{n\pi/2} - \frac{(1)(-\cos n\pi x/2)}{(n\pi/2)^2}\right]_{0}^{2}\right\}.$$

$$= \frac{1}{2}\left\{\frac{4}{n\pi}\left[\sin\frac{n\pi x}{2}\right]_{-2}^{0} + \left[\frac{2x}{n\pi}\sin\frac{n\pi x}{2} + \frac{4}{n^2\pi^2}\cos\frac{n\pi x}{2}\right]_{0}^{2}\right\}.$$

$$= \frac{1}{2}\left\{\frac{4}{n\pi}(0) + \left(0 + \frac{4}{n^2\pi^2}\cos n\pi\right) - \left(0 + \frac{4}{n^2\pi^2}\right)\right\}.$$

$$= \frac{2}{\pi^2 n^2}(\cos n\pi - 1) = \frac{2\{(-1)^n - 1\}}{\pi^2 n^2}.$$

এবং $b_n = \frac{1}{2}\int_{-2}^{2} f(x)\sin\frac{n\pi x}{2}\,dx.$

$$= \frac{1}{2}\left\{\int_{-2}^{0} f(x)\sin\frac{n\pi x}{2}\,dx + \int_{0}^{2} f(x)\sin\frac{n\pi x}{2}\,dx\right\}.$$

$$= \frac{1}{2}\left\{\int_{-2}^{0} 2\sin\frac{n\pi x}{2}\,dx + \int_{0}^{2} x\sin\frac{n\pi x}{2}\,dx\right\}.$$

$$= \frac{1}{2}\left\{-2\left[\frac{\cos n\pi x/2}{n\pi/2}\right]_{-2}^{0} + \left[x\frac{(-\cos n\pi x/2)}{n\pi/2} - \frac{(1)(-\sin n\pi x/2)}{(n\pi/2)^2}\right]_{0}^{2}\right\}.$$

$$= \frac{1}{2}\left\{\frac{-4}{n\pi}\left[\cos\frac{n\pi x}{2}\right]_{-2}^{0} + \left[\frac{-2x}{n\pi}\cos\frac{n\pi x}{2} + \frac{4}{\pi^2 n^2}\sin\frac{n\pi x}{2}\right]_{0}^{2}\right\}.$$

$$= \frac{1}{2}\left\{\frac{-4}{n\pi}(1 - \cos n\pi) + \left(\frac{-4}{n\pi}\cos n\pi + 0\right) - 0\right\} = \frac{-2}{n\pi}.$$

$$\therefore (2) \Rightarrow f(x) = \frac{3}{2} + \sum_{n=1}^{\infty}\frac{2\{(-1)^n - 1\}}{\pi^2 n^2}\cos\frac{n\pi x}{2} - \sum_{n=1}^{\infty}\frac{2}{n\pi}\sin\frac{n\pi x}{2}.$$

বা $f(x) = \frac{3}{2} + \frac{2}{\pi^2}\sum_{n=1}^{\infty}\frac{\{(-1)^n - 1\}\cos n\pi x/2}{n^2} - \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\sin n\pi x/2}{n}.$

(ii). দেওয়া আছে (Given that)

$$f(x) = \begin{cases} -x, & -3 < x < 0 \\ x, & 0 < x < 3 \end{cases} \quad \ldots (1)$$

আমরা জানি, $-3 < x < 3$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [We know, in the interval $-3 < x < 3$, the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi x}{3} + b_n\sin\frac{n\pi x}{3}\right) \quad \ldots (2)$$

যখন $a_0 = \frac{1}{3}\int_{-3}^{3} f(x)\,dx = \frac{1}{3}\left\{\int_{-3}^{0} f(x)\,dx + \int_{0}^{3} f(x)\,dx\right\}.$

$$= \frac{1}{3}\left\{-\int_{-3}^{0} x\,dx + \int_{0}^{3} x\,dx\right\}.$$ (1) নং দ্বারা।

$$=\frac{1}{3}\left\{-\left[\frac{x^2}{2}\right]_{-3}^{0}+\left[\frac{x^2}{2}\right]_{0}^{3}\right\}.$$

$$=\frac{1}{3}\left\{-\frac{1}{2}(0-9)+\frac{1}{2}(9-0)\right\}=\frac{1}{3}(9)=3.$$

$$a_n=\frac{1}{3}\int_{-3}^{3} f(x)\cos\frac{n\pi x}{3}\,dx.$$

$$=\frac{1}{3}\left\{\int_{-3}^{0} f(x)\cos\frac{n\pi x}{3}\,dx+\int_{0}^{3} f(x)\cos\frac{n\pi x}{3}\,dx\right\}.$$

$$=\frac{1}{3}\left\{-\int_{-3}^{0} x\cos\frac{n\pi x}{3}\,dx+\int_{0}^{3} x\cos\frac{n\pi x}{3}\,dx\right\}.$$

$$=\frac{1}{3}\left\{-\left[\frac{x\sin n\pi x/3}{n\pi/3}-\frac{(1)(-\cos n\pi x/3)}{(n\pi/3)^2}\right]_{-3}^{0}+\left[\frac{x\sin n\pi x/3}{n\pi/3}-\frac{1(-\cos n\pi x/3)}{(n\pi/3)^2}\right]_{0}^{3}\right\}.$$

$$=\frac{1}{3}\left\{-\left[\frac{x\sin n\pi x/3}{n\pi/3}+\frac{\cos n\pi x/3}{(n\pi/3)^2}\right]_{-3}^{0}+\left[\frac{x\sin n\pi x/3}{n\pi/3}+\frac{\cos n\pi x/3}{(n\pi/3)^2}\right]_{0}^{3}\right\}.$$

$$=\frac{1}{3}\left\{-\left(0+\frac{1}{(n\pi/3)^2}\right)+\left(0+\frac{\cos n\pi}{(n\pi/3)^2}\right)+\left(0+\frac{\cos n\pi}{(n\pi/3)^2}\right)-\left(0+\frac{1}{(n\pi/3)^2}\right)\right\}$$

$$=\frac{3^2.\,2}{3n^2\pi^2}(\cos n\pi-1)=\frac{6\{(-1)^n-1\}}{\pi^2 n^2}.$$

এবং $b_n=\frac{1}{3}\int_{-3}^{3} f(x)\sin\frac{n\pi x}{3}\,dx,$

$$=\frac{1}{3}\left\{\int_{-3}^{0} f(x)\sin\frac{n\pi x}{3}\,dx+\int_{0}^{3} f(x)\sin\frac{n\pi x}{3}\,dx\right\}.$$

$$=\frac{1}{3}\left\{-\int_{-3}^{0} x\sin\frac{n\pi x}{3}\,dx+\int_{0}^{3} x\sin\frac{n\pi x}{3}\,dx\right\}.$$

$$=\frac{1}{3}\left\{-\left[\frac{x(-\cos n\pi x/3)}{n\pi/3}-\frac{1(-\sin n\pi x/3)}{(n\pi/3)^2}\right]_{-3}^{0}+\left[\frac{x(-\cos n\pi x/3)}{n\pi/3}-\frac{1.(-\sin n\pi x/3)}{(n\pi/3)^2}\right]_{0}^{3}\right\}.$$

$$=\frac{1}{3}\left\{-\left[\frac{-x\cos n\pi x/3}{n\pi/3}+\frac{\sin n\pi x/3}{(n\pi/3)^2}\right]_{-3}^{0}+\left[\frac{-x\cos n\pi x/3}{n\pi/3}+\frac{\sin n\pi x/3}{(n\pi/3)^2}\right]_{0}^{3}\right\}.$$

$$=\frac{1}{3}\left\{-(0+0)+\left(\frac{3\cos n\pi}{n\pi/3}+0\right)+\left(\frac{-3\cos n\pi}{n\pi/3}+0\right)-(0+0)\right\}.$$

$$=0.$$

$$\therefore (2)\Rightarrow f(x)=\frac{3}{2}+\sum_{n=1}^{\infty}\frac{6\{(-1)^n-1\}\cos n\pi x/3}{\pi^2 n^2}+\sum_{n=1}^{\infty}0\sin\frac{n\pi x}{3}$$

$$\text{বা } f(x)=\frac{3}{2}+\frac{6}{\pi^2}\left(\frac{-2\cos\pi x/3}{1^2}+0+\frac{-2\cos 3\pi x/3}{3^2}+0+\frac{-2\cos 5\pi x/3}{5^2}+\dots\right).$$

$$\therefore f(x)=\frac{3}{2}-\frac{12}{\pi^2}\left(\frac{\cos\pi x/3}{1^2}+\frac{\cos 3\pi x/3}{3^2}+\frac{\cos 5\pi x/3}{5^2}+\dots\right).$$

4 iii). দেওয়া আছে (Given that)

$$f(x)=\begin{cases}0, \text{ যখন } -2<x<0\\ 1, \text{ যখন } \ \ 0<x<2\end{cases}\quad \dots(1)$$

আমরা জানি, $-2<x<2$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [We know, in the interval $-2<x<2$, the Fourier series is].

$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi x}{2}+b_n\sin\frac{n\pi x}{2}\right)\ \dots(2)$$

$\frac{\pi}{4}=1-\frac{1}{3}+\frac{1}{5}-\dots$

যখন $a_0 = \frac{1}{2}\int_{-2}^{2} f(x)\,dx = \frac{1}{2}\left\{\int_{-2}^{0} f(x)\,dx + \int_{0}^{2} f(x)\,dx\right\}$.

$= \frac{1}{2}\left\{\int_{-2}^{0} 0\,dx + \int_{0}^{2} 1\,dx\right\}$, (1) নং দ্বারা।

$= \frac{1}{2}\left\{0 + [x]_0^2\right\} = \frac{1}{2}(2-0) = 1.$

$a_n = \frac{1}{2}\int_{-2}^{0} f(x)\cos\frac{n\pi x}{2}\,dx,$

$= \frac{1}{2}\left\{\int_{-2}^{0} f(x)\cos\frac{n\pi x}{2}\,dx + \int_{0}^{2} f(x)\cos\frac{n\pi x}{2}\,dx\right\}.$

$= \frac{1}{2}\left\{\int_{-2}^{0} 0\cos\frac{n\pi x}{2}\,dx + \int_{0}^{2} 1\cos\frac{n\pi x}{2}\,dx\right\}$; (1) নং দ্বারা।

$= \frac{1}{2}\left\{0 + \left[\frac{\sin n\pi x/2}{n\pi/2}\right]_0^2\right\} = \frac{1}{2}\cdot\frac{2}{n\pi}(\sin n\pi - 0) = 0.$

এবং $b_n = \frac{1}{2}\int_{-2}^{2} f(x)\sin\frac{n\pi x}{2}\,dx$.

$= \frac{1}{2}\left\{\int_{-2}^{0} f(x)\sin\frac{n\pi x}{2}\,dx + \int_{0}^{2} f(x)\sin\frac{n\pi x}{2}\,dx\right\}.$

$= \frac{1}{2}\left\{\int_{-2}^{0} 0.\sin\frac{n\pi x}{2}\,dx + \int_{0}^{2} 1.\sin\frac{n\pi x}{2}\,dx\right\}$, (1) নং দ্বারা।

$= \frac{1}{2}\left\{0 + \left[\frac{-\cos n\pi x/2}{n\pi/2}\right]_0^2\right\} = \frac{1}{2}\cdot\frac{2}{n\pi}(-\cos n\pi + 1).$

$= \frac{1-(-1)^n}{n\pi}.$

$\therefore$ (2) $\Rightarrow f(x) = \frac{1}{2} + \sum_{n=1}^{\infty}\left[0 + \frac{\{1-(-1)^n\}\sin n\pi x/2}{n\pi}\right].$

বা, $f(x) = \frac{1}{2} + \frac{1}{\pi}\left(\frac{2\sin\pi x/2}{1} + 0 + \frac{2\sin 3\pi x/2}{3} + 0 + \frac{2\sin 5\pi x/2}{5} + \ldots\right).$

$\therefore f(x) = \frac{1}{2} + \frac{2}{\pi}\left(\frac{\sin\pi x/2}{1} + \frac{\sin 3\pi x/2}{3} + \frac{\sin 5\pi x/2}{5} + \ldots\right)$ .... (3)

দ্বিতীয় অংশ : এখন (3) নং এ $x = 1$ বসাইয়া পাই, [Now in (3), we put $x = 1$ then we get]

$$f(1) = \frac{1}{2} + \frac{2}{\pi}\left(\frac{\sin\pi/2}{1} + \frac{\sin 3\pi/2}{3} + \frac{\sin 5\pi/2}{5} + \ldots\right).$$

বা $1 = \frac{1}{2} + \frac{2}{\pi}\left(1 - \frac{1}{3} + \frac{1}{5} - \ldots\right)$; যেহেতু $0 < x < 2$ ব্যবধিতে $f(x) = 1$

বা $1 - \frac{1}{2} = \frac{2}{\pi}\left(1 - \frac{1}{3} + \frac{1}{5} - \ldots\right).$

বা $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \ldots$, প্রমাণিত।

(iv). দেওয়া আছে (Given that)

$$f(x) = \begin{cases} -1, \text{ যখন } -3 < x < 0 \\ 0, \text{ যখন } x = 0 \\ 1, \text{ যখন } 0 < x < 3 \end{cases} \quad \ldots (1)$$

আমরা জানি, $-3 < x < 3$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল [We know, in the interval $-3 < x < 3$, the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n\cos\frac{n\pi x}{3} + b_n\sin\frac{n\pi x}{3}\right) \quad .... (2)$$

যখন $a_0 = \frac{1}{3}\int_{-3}^{3} f(x)\,dx = \frac{1}{3}\left\{\int_{-3}^{0} f(x)\,dx + \int_{0}^{3} f(x)\,dx\right\}.$

$= \frac{1}{3}\left\{-\int_{-3}^{0} dx + \int_{0}^{3} 1\,dx\right\}$; (1) নং দ্বারা।

$= \frac{1}{3}\left\{-[x]_{-3}^{0} + [x]_0^3\right\} = \frac{1}{3}\{-(0+3) + (3-0)\} = 0.$

$a_n = \frac{1}{3}\int_{-3}^{3} f(x)\cos\frac{n\pi x}{3}\,dx,$

$= \frac{1}{3}\left\{\int_{-3}^{0} f(x)\cos\frac{n\pi x}{3}\,dx + \int_{0}^{3} f(x)\cos\frac{n\pi x}{3}\,dx\right\}.$

$= \frac{1}{3}\left\{-\int_{-3}^{0} 1.\cos\frac{n\pi x}{3}\,dx + \int_{0}^{3} 1.\cos\frac{n\pi x}{3}\right\}$; (1) নং দ্বারা।

$$=\frac{1}{3}\left\{-\left[\frac{\sin n\pi x/3}{n\pi/3}\right]_{-3}^{0}+\left[\frac{\sin n\pi x/3}{n\pi/3}\right]_{0}^{3}\right\}.$$

$$=\frac{1}{3}\left\{-\frac{3}{n\pi}(0-\sin n\pi)+\frac{3}{n\pi}(\sin n\pi-0)\right\}=0,$$

এবং $b_n=\frac{1}{3}\int_{-3}^{3} f(x)\sin\frac{n\pi x}{3}dx,$

$$=\frac{1}{3}\left\{\int_{-3}^{0} f(x)\sin\frac{n\pi x}{3}dx+\int_{0}^{3} f(x)\sin\frac{n\pi x}{3}dx\right\}.$$

$$=\frac{1}{3}\left\{-\int_{-3}^{0} 1.\sin\frac{n\pi x}{3}dx+\int_{0}^{3} 1.\sin\frac{n\pi x}{3}dx\right\}.\ (1)\text{ নং দ্বারা।}$$

$$=\frac{1}{3}\left\{\left[\frac{\cos n\pi x/3}{n\pi/3}\right]_{-3}^{0}-\left[\frac{\cos n\pi x/3}{n\pi/3}\right]_{0}^{3}\right\}.$$

$$=\frac{1}{3}\left\{\frac{3}{n\pi}(1-\cos n\pi)-\frac{3}{n\pi}(\cos n\pi-1)\right\}.$$

$$=\frac{3.2}{3n\pi}(1-\cos n\pi)=\frac{2\{1-(-1)^n\}}{n\pi}.$$

$$\therefore (2)\Rightarrow f(x)=0+\sum_{n=1}^{\infty}\frac{2\{1-(-1)^n\}\sin n\pi x/3}{\pi n}.$$

$$\text{বা } f(x)=\frac{2}{\pi}\left(\frac{2\sin\pi x/3}{1}+0+\frac{2\sin 3\pi x/3}{3}+0+\frac{2\sin 5\pi x/3}{5}+\dots\right)$$

$$\therefore f(x)=\frac{4}{\pi}\left(\frac{\sin\pi x/3}{1}+\frac{\sin 3\pi x/3}{3}+\frac{\sin 5\pi x/3}{5}+\dots\right).$$

5(i). দেওয়া আছে (Given that)

$$f(x)=\begin{cases}\pi+x, & -\pi<x<0\\ \pi-x, & 0<x<\pi\end{cases}\quad\dots(1)$$

আমরা জানি, $-\pi<x<\pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [We know, in he interval $-\pi<x<\pi$, the Fourier series is].

$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n\cos nx+b_n\sin nx),\ \dots(2)$$

যখন $a_0=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx=\frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\,dx+\int_{0}^{\pi} f(x)\,dx\right\}.$

$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0}(\pi+x)\,dx+\int_{0}^{\pi}(\pi-x)\,dx\right\};\ (1)\text{ নং দ্বারা।}$$

$$=\frac{1}{\pi}\left\{\left[\pi x+\frac{x^2}{2}\right]_{-\pi}^{0}+\left[\pi x-\frac{x^2}{2}\right]_{0}^{\pi}\right\}.$$

$$=\frac{1}{\pi}\left\{0-\left(-\pi^2+\frac{\pi^2}{2}\right)+\left(\pi^2-\frac{\pi^2}{2}\right)-0\right\}=\frac{1}{\pi}\left(\frac{\pi^2}{2}+\frac{\pi^2}{2}\right)=\pi.$$

$$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx,$$

$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\cos nx\,dx+\int_{0}^{\pi} f(x)\cos nx\,dx\right\}.$$

$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0}(\pi+x)\cos nx\,dx+\int_{0}^{\pi}(\pi-x)\cos nx\,dx\right\}.$$

$$=\frac{1}{\pi}\left\{\left[\frac{(\pi+x)\sin nx}{n}-\frac{(1)(-\cos nx)}{n^2}\right]_{-\pi}^{0}+\left[\frac{(\pi-x)\sin nx}{n}-\frac{(-1)(-\cos nx)}{n^2}\right]_{0}^{\pi}\right\}.$$

$$=\frac{1}{\pi}\left\{\left[\frac{(\pi+x)\sin nx}{n}+\frac{\cos nx}{n^2}\right]_{-\pi}^{0}+\left[\frac{(\pi-x)\sin nx}{n}-\frac{\cos nx}{n^2}\right]_{0}^{\pi}\right\}.$$

$$=\frac{1}{\pi}\left\{\left(0+\frac{1}{n^2}\right)-\left(0+\frac{\cos n\pi}{n^2}\right)+\left(0-\frac{\cos n\pi}{n^2}\right)-\left(0-\frac{1}{n^2}\right)\right\}$$

$$=\frac{2}{\pi n^2}(1-\cos n\pi)=\frac{2\{1-(-1)^n\}}{\pi n^2}.$$

এবং $b_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx,$

$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\sin nx\,dx+\int_{0}^{\pi} f(x)\sin nx.dx\right\}.$$

$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0}(\pi+x)\ \sin nx\ dx+\int_{0}^{\pi}(\pi-x)\ \sin nx\ dx\right\};\ (1)\ \text{নং দ্বারা।}$$

$$=\frac{1}{\pi}\left\{\left[\frac{(\pi+x)\ (-\cos nx)}{n}-\frac{(1)\ (-\sin nx)}{n^2}\right]_{-\pi}^{0}\right.$$

$$\left.+\left[\frac{(\pi-x)\ (-\cos nx)}{n}-\frac{(-1)\ (-\sin nx)}{n^2}\right]_{0}^{\pi}\right\}.$$

$$=\frac{1}{\pi}\left\{\left[\frac{-(\pi+x)\cos nx}{n}+\frac{\sin nx}{n^2}\right]_{-\pi}^{0}\right.$$

$$\left.+\left[\frac{-(\pi-x)\cos nx}{n}-\frac{\sin nx}{n^2}\right]_{0}^{\pi}\right\}.$$

$$=\frac{1}{\pi}\left\{\left(\frac{-\pi}{n}+0\right)+0+(0)+\left(\frac{\pi}{n}-0\right)\right\}=\frac{1}{\pi}\left(\frac{-\pi}{n}+\frac{\pi}{n}\right)=0.$$

$$\therefore (2)\Rightarrow f(x)=\frac{\pi}{2}+\frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\{1-(-1)^n\}\cos nx}{n^2}+0.$$

$$\text{বা } f(x)=\frac{\pi}{2}+\frac{2}{\pi}\left(\frac{2\cos x}{1^2}+0+\frac{2\cos 3x}{3^2}+0+\frac{2\cos 5x}{5^2}+\dots\right).$$

$$\text{বা } f(x)=\frac{\pi}{2}+\frac{4}{\pi}\left(\frac{\cos x}{1^2}+\frac{\cos 3x}{3^2}+\frac{\cos 5x}{5^2}+\dots\right).$$

(ii). দেওয়া আছে (given that)

$$f(x)=\begin{cases}1+x/\pi, & \text{যখন } -\pi\le x<0\\ 1-x/\pi, & \text{যখন } \ 0\le x\le\pi\end{cases}\quad \dots (1)$$

আমরা জানি, $-\pi\le x\le\pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [We know, in the interval $-\pi\le x\le x$, the Fourier series is]

$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}(a_n\cos nx+b_n\sin nx),\ \dots (2)$$

$$\text{যখন } a_0=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\ dx.$$

$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0}f(x)\ dx+\int_{0}^{\pi}f(x)\ dx\right\}.$$

ফ্যুরিয়ার ধারা

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$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0}\left(1+\frac{x}{\pi}\right)dx+\int_{0}^{\pi}\left(1-\frac{x}{\pi}\right)dx\right\};\ (1)\ \text{নং দ্বারা।}$$

$$=\frac{1}{\pi}\left\{\left[x+\frac{x^2}{2\pi}\right]_{-\pi}^{0}+\left[x-\frac{x^2}{2\pi}\right]_{0}^{\pi}\right\}.$$

$$=\frac{1}{\pi}\left\{0-\left(-\pi+\frac{\pi^2}{2\pi}\right)+\left(\pi-\frac{\pi^2}{2\pi}\right)-0\right\}.$$

$$=\frac{2}{\pi}\left(\pi-\frac{\pi^2}{2\pi}\right)=\frac{2}{\pi}\left(\pi-\frac{\pi}{2}\right)=\frac{2}{\pi}\cdot\frac{\pi}{2}=1.$$

$$a_n=\frac{1}{\pi}\int_{-\pi}^{\pi}f(x)\cos nx\ dx.$$

$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0}f(x)\cos nx\ dx+\int_{0}^{\pi}f(x)\cos nx\ dx\right\}.$$

$$=\frac{1}{2}\left\{\int_{-\pi}^{0}\left(1+\frac{x}{\pi}\right)\cos nx\ dx+\int_{0}^{\pi}\left(1-\frac{x}{\pi}\right)\cos nx\ dx\right\}.$$

(1) নং দ্বারা।

$$=\frac{1}{\pi}\left\{\left[\left(1+\frac{x}{\pi}\right)\frac{\sin nx}{n}-\left(\frac{1}{\pi}\right)\frac{(-\cos nx)}{n^2}\right]_{-\pi}^{0}\right.$$

$$\left.+\left[\left(1-\frac{x}{\pi}\right)\frac{\sin nx}{n}-\left(-\frac{1}{\pi}\right)\frac{(-\cos nx)}{n^2}\right]_{0}^{\pi}\right\}.$$

$$=\frac{1}{\pi}\left\{\left[\left(1+\frac{x}{\pi}\right)\frac{\sin nx}{n}+\frac{\cos nx}{\pi n^2}\right]_{-\pi}^{0}\right.$$

$$\left.+\left[\left(1-\frac{x}{\pi}\right)\frac{\sin nx}{n}-\frac{\cos nx}{\pi n^2}\right]_{0}^{\pi}\right\}.$$

$$=\frac{1}{\pi}\left\{\left(0+\frac{1}{\pi n^2}\right)-\left(0+\frac{\cos n\pi}{\pi n^2}\right)\right.$$

$$\left.+\left(0-\frac{\cos n\pi}{\pi n^2}\right)-\left(0-\frac{1}{\pi n^2}\right)\right\}$$

$$= \frac{1}{\pi}\left(\frac{2}{\pi n^2} - \frac{2\cos n\pi}{\pi n^2}\right) = \frac{2}{\pi^2 n^2}(1-\cos n\pi).$$

$$= \frac{2\{1-(-1)^n\}}{\pi^2 n^2}.$$

এবং $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\, dx$,

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\sin nx\, dx + \int_{0}^{\pi} f(x)\sin nx\, dx\right\}.$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0}\left(1+\frac{x}{\pi}\right)\sin nx\, dx + \int_{0}^{\pi}\left(1-\frac{x}{\pi}\right)\sin nx\, dx\right\}.$$

$$= \frac{1}{\pi}\left\{\left[\left(1+\frac{x}{\pi}\right)\frac{(-\cos nx)}{n} - \left(\frac{1}{\pi}\right)\frac{(-\sin nx)}{n^2}\right]_{-\pi}^{0} + \left[\left(1-\frac{x}{\pi}\right)\frac{(-\cos nx)}{n} - \left(-\frac{1}{\pi}\right)\frac{(-\sin nx)}{n^2}\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{\left[-\left(1+\frac{x}{\pi}\right)\frac{\cos nx}{n} + \frac{\sin nx}{\pi n^2}\right]_{-\pi}^{0} + \left[-\left(1-\frac{x}{\pi}\right)\frac{\cos nx}{n} - \frac{\sin nx}{\pi n^2}\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{-(1+0)\frac{1}{n} + 0 + 0 + (1-0)\frac{1}{n}\right\} = \frac{1}{\pi}\left(-\frac{1}{n}+\frac{1}{n}\right) = 0.$$

$$\therefore (2) \Rightarrow f(x) = \frac{1}{2} + \sum_{n=1}^{\infty}\frac{2\{1-(-1)^n\}\cos nx}{\pi^2 n^2} + 0,$$

বা $f(x) = \frac{1}{2} + \frac{2}{\pi^2}\sum_{n=1}^{\infty}\frac{\{1-(-1)^n\}\cos nx}{n^2}.$

বা $f(x) = \frac{1}{2} + \frac{2}{\pi^2}\left(\frac{2\cos x}{1^2} + 0 + \frac{2\cos 3x}{3^2} + 0 + \frac{2\cos 5x}{5^2} + \dots\right).$

বা $f(x) = \frac{1}{2} + \frac{4}{\pi^2}\left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots\right)$, ... (3)

দ্বিতীয় অংশ ঃ যেহেতু $x = 0$ বিন্দুতে $f(x)$ ফাংশন অবিচ্ছিন্ন, কাজেই প্রদত্ত শর্ত হইতে পাই, $f(0) = 1 - \frac{0}{\pi} = 1$. [Since the tunction $f(x)$ is continuous at the point $x = 0$, so from the given condition, we get $f(0) = 1 - \frac{0}{\pi} = 1$].

এখন (3) নং এ $x = 0$ বসাইয়া পাই, [Now put $x = 0$ in (3), we get]

$$f(0) = \frac{1}{2} + \frac{4}{\pi^2}\left(\frac{1}{1^2} + \frac{1}{3^2} + \frac{1}{5^2} + \dots\right).$$

বা $1 = \frac{1}{2} + \frac{4}{\pi^2}\left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots\right).$

বা $\frac{1}{2} = \frac{4}{\pi^2}\left(1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots\right).$

$$\therefore \frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

(iii). দেওয়া আছে (Given that)

$$f(x) = \begin{cases} \pi - x, & \text{যখন } -\pi < x < 0 \\ x - \pi, & \text{যখন } 0 < x < \pi \end{cases} \quad \dots (1)$$

আমরা জানি, $-\pi < x < \pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [We know, in the interval $-\pi < x < \pi$ the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n\cos nx + b_n\sin nx), \dots (2)$$

যখন $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\, dx = \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\, dx + \int_{0}^{\pi} f(x)\, dx\right\}.$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0}(\pi - x)\, dx + \int_{0}^{\pi}(x-\pi)\, dx\right\}. \text{ (1) নং দ্বারা।}$$

$$= \frac{1}{\pi}\left\{\left[\pi x - \frac{x^2}{2}\right]_{-\pi}^{0} + \left[\frac{x^2}{2} - \pi x\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{0 - \left(-\pi^2 - \frac{\pi^2}{2}\right) + \left(\frac{\pi^2}{2} - \pi^2\right) - 0\right\}.$$

$$= \frac{1}{\pi}(\pi^2) = \pi.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx.$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\cos nx\,dx + \int_{0}^{\pi} f(x)\cos nx\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} (\pi-x)\cos nx\,dx + \int_{0}^{\pi} (x-\pi)\cos nx\,dx\right\}, \text{ (1) নং দ্বারা।}$$

$$= \frac{1}{\pi}\left\{\left[(\pi-x)\frac{\sin nx}{n} - (-1)\frac{(-\cos nx)}{n^2}\right]_{-\pi}^{0} + \left[(x-\pi)\frac{\sin nx}{n} - (1)\frac{(-\cos nx)}{n^2}\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{\left[(\pi-x)\frac{\sin nx}{n} - \frac{\cos nx}{n^2}\right]_{-\pi}^{0} + \left[(x-\pi)\frac{\sin nx}{n} + \frac{\cos nx}{n^2}\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{\left(0-\frac{1}{n^2}\right) - \left(0-\frac{\cos n\pi}{n^2}\right) + \left(0+\frac{\cos n\pi}{n^2}\right) - \left(0+\frac{1}{n^2}\right)\right\}.$$

$$= \frac{2}{\pi n^2}(\cos n\pi - 1) = \frac{2\{(-1)^n - 1\}}{\pi n^2}.$$

এবং $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx.$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\sin nx\,dx + \int_{0}^{\pi} f(x)\sin nx\,dx\right\}.$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} (\pi-x)\sin nx\,dx + \int_{0}^{\pi} (x-\pi)\sin nx\,dx\right\}, \text{ (1) নং দ্বারা।}$$

$$= \frac{1}{\pi}\left\{\left[\frac{(\pi-x)(-\cos nx)}{n} - \frac{(-1)-(-\sin nx)}{n^2}\right] + \left[\frac{(x-\pi)(-\cos nx)}{n} - \frac{(1)(-\sin nx)}{n^2}\right]\right\}$$

$$= \frac{1}{\pi}\left\{-\left[\frac{(\pi-x)\cos nx}{n} - \frac{\sin nx}{n^2}\right]_{-\pi}^{0} + \left[\frac{-(x-\pi)\cos nx}{n} + \frac{\sin nx}{n^2}\right]_{0}^{\pi}\right\}.$$

$$= \frac{1}{\pi}\left\{-\frac{\pi}{n} + \frac{2\pi\cos n\pi}{n} + 0 + \left(\frac{-\pi}{n}\right)\right\} = \frac{1}{\pi}\left(\frac{2\pi\cos n\pi}{n} - \frac{2\pi}{n}\right).$$

$$= \frac{2\pi}{\pi n}(\cos n\pi - 1) = \frac{2\{(-1)^n - 1\}}{n}.$$

$$\therefore (2) \Rightarrow f(x) = \frac{\pi}{2} + \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{\{(-1)^n - 1\}\cos nx}{n^2} + 2\sum_{n=1}^{\infty}\frac{\{(-1)^n - 1\}\sin nx}{n}.$$

$$\text{বা } f(x) = \frac{\pi}{2} + \frac{2}{\pi}\left(\frac{-2\cos x}{1^2} + 0 + \frac{-2\cos 3x}{3^2} + 0 + \frac{-2\cos 5x}{5^2} + \dots\right) + 2\left(\frac{-2\sin x}{1} + 0 + \frac{-2\sin 3x}{3} + 0 + \frac{-2\sin 5x}{5} + \dots\right).$$

$$\therefore f(x) = \frac{\pi}{2} - \frac{4}{\pi}\left(\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots\right) - 4\left(\frac{\sin x}{1} + \frac{\sin 3x}{3} + \frac{\sin 5x}{5} + \dots\right).$$

6(i). দেওয়া আছে (Given that) $f(x) = x\cos x$

$$f(x) = x\cos x,\ -\pi < x < \pi$$

$$\therefore\ f(-x) = -x\cos(-x) = -x\cos x = -f(x).$$

$\Rightarrow f(x)$ অযুগ্ম ফাংশন। কাজেই ইহার ফ্যুরিয়ার ধারা হইল, [$\Rightarrow f(x)$ is an odd function, so its Fourier series is]

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx, \dots (1)$$

$$x\cos x = -\frac{\sin x}{2} + 2\sum_{n=2}^{\infty}\frac{(-1)^n n\sin nx}{n^2-1}$$

যখন $b_n = \frac{2}{\pi}\int_0^{\pi} f(x) \sin nx\, dx$,

$$= \frac{2}{\pi}\int_0^{\pi} x \cos x \sin nx\, dx, \ldots (2)$$

$$= \frac{1}{\pi}\int_0^{\pi} x\{\sin(n+1)x + \sin(n-1)x\}\, dx,$$

$$= \frac{1}{\pi}\Bigg[x\left\{\frac{-\cos(n+1)x}{(n+1)} - \frac{\cos(n-1)x}{n-1}\right\}$$

$$- 1\left\{\frac{-\sin(n+1)x}{(n+1)^2} - \frac{\sin(n-1)x}{(n-1)^2}\right\}\Bigg]_0^{\pi}, n \neq 1$$

$$= \frac{1}{\pi}\Bigg[-x\left\{\frac{\cos(n+1)x}{n+1} + \frac{\cos(n-1)x}{n-1}\right\}$$

$$+ \frac{\sin(n+1)x}{(n+1)^2} + \frac{\sin(n-1)x}{(n-1)^2}\Bigg]_0^{\pi}$$

$$= \frac{1}{\pi}\left[-\pi\left\{\frac{\cos(n+1)\pi}{n+1} + \frac{\cos(n-1)\pi}{n-1}\right\} + 0\right].$$

$$= -\frac{\cos(\pi+n\pi)}{n+1} - \frac{\cos(\pi-n\pi)}{n-1} = \frac{\cos n\pi}{n+1} + \frac{\cos n\pi}{n-1}.$$

$$b_n = \frac{2n \cos n\pi}{(n-1)(n+1)} = \frac{2n(-1)^n}{n^2-1}, n = 2, 3, 4, \ldots$$

এখন (2) নং এ $n = 1$ বসাইয়া পাই, [Now we put $n = 1$ in (2) then we get]

$$b_1 = \frac{2}{\pi}\int_0^{\pi} x \cos x \sin x\, dx,$$

$$= \frac{1}{\pi}\int_0^{\pi} x \sin 2x\, dx.$$

$$= \frac{1}{\pi}\left[x\frac{(-\cos 2x)}{2} - (1).\frac{(-\sin 2x)}{2^2}\right]_0^{\pi}.$$

$$= \frac{1}{\pi}\left[-\frac{x\cos 2x}{2} + \frac{\sin 2x}{4}\right]_0^{\pi}.$$

$$= \frac{1}{\pi}\left\{\frac{-\pi\cos 2\pi}{2} + 0\right\} = -\frac{1}{2}.$$

$$\therefore (1) \Rightarrow f(x) = b_1 \sin x + \sum_{n=2}^{\infty} b_n \sin nx.$$

$$\therefore x\cos x = -\frac{1}{2}\sin x + 2\sum_{n=2}^{\infty}\frac{n(-1)^n}{n^2-1}\sin nx.$$ প্রমাণিত।

(ii). দেওয়া আছে (Given that)

$$f(x) = \begin{cases} 0, \text{ যখন } -\pi < x < 0 \\ \sin x, \text{ যখন } 0 < x < \pi \end{cases} \ldots (1)$$

আমরা জানি, $-\pi < x < \pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল [We know, in the interval $-\pi < x < \pi$ the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}(a_n \cos nx + b_n \sin nx), \ldots (2)$$

যখন $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\, dx = \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\, dx + \int_0^{\pi} f(x)\, dx\right\}$,

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\, dx + \int_0^{\pi} \sin x\, dx\right\}, \text{ (1) নং দ্বারা।}$$

$$= \frac{1}{\pi}\left\{0 - [\cos x]_0^{\pi}\right\} = \frac{1}{\pi}\{-(\cos\pi - \cos 0)\} = \frac{2}{\pi},$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x) \cos nx\, dx,$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\cos nx\, dx + \int_0^{\pi} f(x) \cos nx\, dx\right\},$$

$$= \frac{1}{\pi}\left\{\int_{-\pi}^{0} 0 \cos nx\, dx + \int_0^{\pi} \sin x \cos nx\, dx\right\} \ldots (3)$$

$$= \frac{1}{\pi}\left[0 + \frac{1}{2}\int_0^{\pi}\{\sin(n+1)x - \sin(n-1)x\}\, dx\right].$$

$$=\frac{1}{2\pi}\left[\frac{-\cos(n+1)x}{n+1}+\frac{\cos(n-1)x}{n-1}\right]_0^\pi \ ; n\neq 1.$$

$$=\frac{1}{2\pi}\left\{\frac{-\cos(n+1)\pi}{n+1}+\frac{\cos(n-1)\pi}{n-1}+\frac{1}{n+1}-\frac{1}{n-1}\right\}.$$

$$=\frac{1}{2\pi}\left\{\frac{-\cos(\pi+n\pi)}{n+1}+\frac{\cos(\pi-n\pi)}{n-1}+\frac{1}{n+1}-\frac{1}{n-1}\right\}.$$

$$=\frac{1}{2\pi}\left\{\frac{\cos n\pi}{n+1}-\frac{\cos n\pi}{n-1}+\frac{1}{n+1}-\frac{1}{n-1}\right\}.$$

$$=\frac{1}{2\pi}\left\{\frac{2\cos n\pi}{n^2-1}+\frac{-2}{n^2-1}\right\}=\frac{2}{2\pi(n^2-1)}(\cos n\pi-1).$$

$$=\frac{(-1)^n-1}{\pi(n^2-1)}\ ; n=2,3,4,\ldots$$

এখন (3) নং এ $n=1$ বসাইয়া পাই, [Now we put $n=1$ in (3) then we get]

$$a_1=\frac{1}{\pi}\int_0^\pi \sin x\cos x\,dx=\frac{1}{2\pi}\int_0^\pi \sin 2x\,dx,$$

$$=\frac{1}{2\pi}\left[\frac{-\cos 2x}{2}\right]_0^\pi=-\frac{1}{4\pi}(\cos 2\pi-1)$$

$$=0.$$

এবং $b_n=\frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx,$

$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0} f(x)\sin nx\,dx+\int_0^\pi f(x)\sin nx\,dx\right\}.$$

$$=\frac{1}{\pi}\left\{\int_{-\pi}^{0} 0\sin nx\,dx+\int_0^\pi \sin x\sin nx\,dx\right\};\ \ldots (4)$$

$$=\frac{1}{\pi}\left[0+\frac{1}{2}\int_0^\pi\{\cos(n-1)x-\cos(n+1)x\}\,dx\right],$$

$$=\frac{1}{2\pi}\left[\frac{\sin(n-1)x}{n-1}-\frac{\sin(n+1)x}{n+1}\right]_0^\pi,\ n\neq 1,$$

$=0$, যেহেতু $\sin(n-1)\pi=0$, $\sin(n+1)\pi=0$.

(4) নং এ $n=1$ বসাইয়া পাই, [We put $n=1$ in (4) then we get]

$$b_1=\frac{1}{\pi}\int_0^\pi \sin x\sin x\,dx=\frac{1}{2\pi}\int_0^\pi(1-\cos 2x)\,dx.$$

$$=\frac{1}{2\pi}\left[x-\frac{\sin 2x}{2}\right]_0^\pi=\frac{1}{2\pi}(\pi-0)=\frac{1}{2}.$$

$$\therefore (2)\Rightarrow f(x)=\frac{a_0}{2}+a_1\cos x+\sum_{n=2}^{\infty}a_n\cos nx+b_1\sin x+\sum_{n=2}^{\infty}b_n\sin nx.$$

বা $f(x)=\frac{1}{2}\cdot\frac{2}{\pi}+0+\frac{1}{\pi}\sum_{n=2}^{\infty}\frac{\{(-1)^n-1\}\cos nx}{(n-1)(n+1)}+\frac{1}{2}\sin x+0,$

বা $f(x)=\frac{1}{\pi}+\frac{1}{\pi}\left(0-\frac{2\cos 3x}{2.4}+0-\frac{2\cos 5x}{4.6}+0-\frac{2\cos 7x}{6.8}+\ldots\right)+\frac{\sin x}{2}$

$$\therefore f(x)=\frac{1}{\pi}-\frac{2}{\pi}\left(\frac{\cos 3x}{2.4}+\frac{\cos 5x}{4.6}+\frac{\cos 7x}{6.8}+\ldots\right)+\frac{\sin x}{2}.$$

(iii)(a). দেওয়া আছে (Given that)

$\therefore f(x)=\cos\alpha x,$

$\because\therefore f(-x)=\cos(-\alpha x)=\cos\alpha x=f(x)$

$\Rightarrow f(x)$ যুগ্ম ফাংশন, কাজেই ইহার ফ্যুরিয়ার ধারা হইল, [$\Rightarrow f(x)$ is an even function, so its Fourier series is]

$$f(x)=\frac{a_0}{2}+\sum_{n=1}^{\infty}a_n\cos nx,\ \ldots (1)$$

যখন $a_0=\frac{2}{\pi}\int_0^\pi f(x)\,dx$ এবং $a_n=\frac{2}{\pi}\int_0^\pi f(x)\cos nx\,dx.$

এখন $a_0=\frac{2}{\pi}\int_0^\pi \cos\alpha x\,dx=\frac{2}{\pi}\left[\frac{\sin\alpha x}{\alpha}\right]_0^\pi.$

$$=\frac{2}{\pi\alpha}(\sin\alpha\pi-0)=\frac{2\sin\alpha\pi}{\alpha\pi}.$$

এবং $a_n = \frac{2}{\pi}\int_0^\pi \cos\alpha x.\cos nx\,dx.$

$$= \frac{1}{\pi}\int_0^\pi \{\cos(\alpha-n)x + \cos(\alpha+n)x\}\,dx,$$

$$= \frac{1}{\pi}\left[\frac{\sin(\alpha-n)x}{\alpha-n}+\frac{\sin(\alpha+n)x}{\alpha+n}\right]_0^\pi .$$

$$= \frac{1}{\pi}\left\{\frac{\sin(\alpha-n)\pi}{\alpha-n}+\frac{\sin(\alpha+n)\pi}{\alpha+n}-0\right\}.$$

$$= \frac{1}{\pi}\left\{\frac{\sin(\alpha\pi-n\pi)}{\alpha-n}+\frac{\sin(\alpha\pi+n\pi)}{\alpha+n}\right\}.$$

$$= \frac{1}{\pi}\left(\frac{\sin\alpha\pi\cos n\pi-\cos\alpha\pi\sin n\pi}{\alpha-n} + \frac{\sin\alpha\pi\cos n\pi+\cos\alpha\pi\sin n\pi}{\alpha+n}\right).$$

$$= \frac{1}{\pi}\left(\frac{\sin\alpha\pi\cos n\pi-0}{\alpha-n}+\frac{\sin\alpha\pi\cos n\pi}{\alpha+n}\right).$$

যেহেতু $\sin n\pi = 0$.

$$= \frac{\sin\alpha\pi.\cos n\pi}{\pi}\left(\frac{1}{\alpha-n}+\frac{1}{\alpha+n}\right) = \frac{2\alpha.\sin\alpha\pi.\cos n\pi}{\pi(\alpha^2-n^2)}.$$

$$= \frac{2(-1)^n\alpha.\sin\alpha\pi}{\pi(\alpha^2-n^2)}.$$

∴ (1) ⇒ $f(x) = \frac{2\sin\alpha\pi}{2\alpha\pi}+\sum_{n=1}^{\infty}\frac{2(-1)^n\alpha\sin\alpha\pi\cos nx}{\pi(\alpha^2-n^2)}.$

বা $f(x) = \frac{\sin\alpha\pi}{\alpha\pi}+\frac{2\alpha\sin\alpha\pi}{\pi}\sum_{n=1}^{\infty}\frac{(-1)^n\cos nx}{\alpha^2-n^2}.$

বা $\cos\alpha x = \frac{\sin\alpha\pi}{\alpha\pi}+\frac{2\alpha\sin\alpha\pi}{\pi}\left(\frac{-\cos x}{\alpha^2-1^2}+\frac{\cos 2x}{\alpha^2-2^2}-\frac{\cos 3x}{\alpha^2-3^2}+\cdots\right).$

∴ $\cos\alpha x = \frac{\sin\alpha\pi}{\alpha\pi}-\frac{2\alpha\sin\alpha\pi}{\pi}\left(\frac{\cos x}{\alpha^2-1^2}-\frac{\cos 2x}{\alpha^2-2^2}+\frac{\cos 3x}{\alpha^2-3^2}-\cdots\right)$ ... (2)

(b). এখন (2) নং এ $x = \pi$ বসাইয়া পাই, [Now we put $x = \pi$ is (2) then we get]

$$\cos\alpha\pi = \frac{\sin\alpha\pi}{\alpha\pi}-\frac{2\alpha\sin\alpha\pi}{\pi}\left(\frac{-1}{\alpha^2-1^2}-\frac{1}{\alpha^2-2^2}-\frac{1}{\alpha^2-3^2}-\cdots\right).$$

বা $\cos\alpha\pi = \frac{\sin\alpha\pi}{\pi}\left(\frac{1}{\alpha}+\frac{2\alpha}{\alpha^2-1^2}+\frac{2\alpha}{\alpha^2-2^2}+\frac{2\alpha}{\alpha^2-3^2}+\cdots\right).$

$$\frac{\pi\cos\alpha\pi}{\sin\alpha\pi} = \frac{1}{\alpha}+\frac{2\alpha}{\alpha^2-1^2}+\frac{2\alpha}{\alpha^2-2^2}+\frac{2\alpha}{\alpha^2-3^2}+\cdots.$$

বা $\frac{\pi\cos\alpha\pi}{\sin\alpha\pi}-\frac{1}{\alpha} = \frac{2\alpha}{\alpha^2-1^2}+\frac{2\alpha}{\alpha^2-2^2}+\frac{2\alpha}{\alpha^2-3^2}+\cdots.$

উভয় পক্ষকে 0 এবং x সীমার মধ্যে α-এর সম্পর্কে ইনটিগ্রেট করিয়া পাই, [Integrating both sides w. r. to α between the limits 0 and x then we get]

$$\int_0^x\left(\frac{\pi\cos\alpha\pi}{\sin\alpha\pi}-\frac{1}{\alpha}\right)d\alpha = \int_0^x\frac{2\alpha\,d\alpha}{\alpha^2-1^2}+\int_0^x\frac{2\alpha\,d\alpha}{\alpha^2-2^2}+\int_0^x\frac{2\alpha\,d\alpha}{\alpha^2-3^2}+\cdots.$$

বা $\left[\log\sin\alpha\pi-\log\alpha\right]_0^x = \left[\log(\alpha^2-1^2)\right]_0^x+\left[\log(\alpha^2-2^2)\right]_0^x+\left[\log(\alpha^2-3^2)\right]_0^x+\cdots.$

বা $\left[\log\frac{\sin\alpha\pi}{\alpha}\right]_0^x = \{\log(x^2-1^2)-\log(-1^2)\}+\{\log(x^2-2^2)-\log(-2^2)\}+\{\log(x^2-3^2)-\log(-3^2)\}+\cdots$, ... (3)

এখন $\left[\log\frac{\sin\alpha\pi}{\alpha}\right]_0^x = \log\frac{\sin x\pi}{x}-\log\left(\lim_{\alpha\to0}\frac{\sin\alpha\pi}{\alpha}\right).$

$$= \log\frac{\sin\pi x}{x}-\log\pi\left(\lim_{\alpha\to0}\frac{\sin\alpha\pi}{\alpha\pi}\right).$$

$$= \log \frac{\sin \pi x}{x} - \log \pi . 1,$$

$$= \log \frac{\sin \pi x}{\pi x}, \ldots (4)$$

(3) নং এবং (4) নং হইতে পাই, [From (3) and (4) we get]

$$\log \frac{\sin \pi x}{\pi x} = \log\left(1 - \frac{x^2}{1^2}\right) + \log\left(1 - \frac{x^2}{2^2}\right) + \log\left(1 - \frac{x^2}{3^2}\right) + \ldots,$$

বা $\log \frac{\sin \pi x}{\pi x} = \log\left\{\left(1 - \frac{x^2}{1^2}\right)\left(1 - \frac{x^2}{2^2}\right)\left(1 - \frac{x^2}{3^2}\right)\ldots\right\},$

বা $\frac{\sin \pi x}{\pi x} = \left(1 - \frac{x^2}{1^2}\right)\left(1 - \frac{x^2}{2^2}\right)\left(1 - \frac{x^2}{3^2}\right)\ldots, \ldots (5)$

এখন (5) নং এ x এর স্থলে x/π বসাইয়া পাই, [Now replacing x by x/π in (5) then we get]

$$\frac{\sin x}{x} = \left(1 - \frac{x^2}{\pi^2}\right)\left(1 - \frac{x^2}{(2\pi)^2}\right)\left(1 - \frac{x^2}{(3\pi)^2}\right)\ldots,$$ প্রমাণিত।

এখন $x = \pi/2$ বসাইয়া পাই, [Now we put $x = \pi/2$ then we get]

$$\frac{\sin\pi/2}{\pi/2} = \left(1 - \frac{\pi^2}{2^2\,\pi^2}\right)\left(1 - \frac{\pi^2}{2^2.2^2\,\pi^2}\right)\left(1 - \frac{\pi^2}{2^2.\,3^2.\,\pi^2}\right)\ldots.$$

বা $\frac{2}{\pi} = \left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{4^2}\right)\left(1 - \frac{1}{6^2}\right)\ldots,$

বা $\frac{2}{\pi} = \left(\frac{1.3}{2.2}\right)\left(\frac{3.5}{4.4}\right)\left(\frac{5.7}{6.6}\right)\ldots$

$\therefore \frac{\pi}{2} = \frac{2.2.4.4.6.6.8.8\ldots}{1.3.3.5.5.7.7.9\ldots}$ প্রমাণিত।

(iv). দেওয়া আছে (Given that)

$$f(x) = \frac{\pi e^x}{2 \sin h\pi}, -\pi < x < \pi.$$

আমরা জানি, $-\pi < x < \pi$ ব্যবধিতে ফ্যুরিয়ার ধারা হইল, [We know, in the interval $-\pi < x < \pi$, the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx), \ldots (1)$$

যখন $a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\,dx,$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} \frac{\pi e^x\,dx}{2\sinh\pi} = \frac{1}{2\sinh\pi}\left[e^x\right]_{-\pi}^{\pi}.$$

$$= \frac{1}{2\sinh\pi}(e^{\pi} - e^{-\pi}) = \frac{\sinh\pi}{\sinh\pi} = 1.$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\,dx,$$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} \frac{\pi e^x}{2\sinh\pi}\cos nx\,dx = \frac{1}{2\sinh\pi}\int_{-\pi}^{\pi} e^x \cos nx\,dx.$$

$$= \frac{1}{2\sinh\pi}\left[\frac{e^x(\cos nx + n\sin nx)}{n^2 + 1^2}\right]_{-\pi}^{\pi}.$$

$$= \frac{1}{2\sinh\pi}\left\{\frac{e^{\pi}(\cos n\pi + 0)}{n^2 + 1^2} - \frac{e^{-\pi}(\cos n\pi + 0)}{n^2 + 1^2}\right\};$$

$$= \frac{\cos n\pi}{2(n^2 + 1^2)\sinh\pi}(e^{\pi} - e^{-\pi}) = \frac{\cos n\pi.\,\sinh\pi}{(n^2 + 1^2)\sinh\pi},$$

$$= \frac{(-1)^n}{(n^2 + 1)}.$$

এবং $b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\,dx,$

$$= \frac{1}{\pi}\int_{-\pi}^{\pi} \frac{\pi e^x.\,\sin nx\,dx}{2\sinh\pi} = \frac{1}{2\sinh\pi}\int_{-\pi}^{\pi} e^x \sin nx\,dx.$$

$$= \frac{1}{2\sinh\pi}\left[\frac{e^x(\sin nx - n\cos nx)}{n^2 + 1}\right]_{-\pi}^{\pi}.$$

$$= \frac{1}{2\sinh\pi}\left\{\frac{e^{\pi}(0 - n\cos n\pi)}{n^2 + 1} - \frac{e^{-\pi}(0 - n\cos n\pi)}{n^2 + 1}\right\}.$$

$$= \frac{-n\cos n\pi}{2(n^2 + 1)\sinh\pi}(e^{\pi} - e^{-\pi}) = \frac{-n\cos n\pi.\,\sinh\pi}{(n^2 + 1)\sinh\pi}$$

$$= \frac{-n(-1)^n}{n^2 + 1}.$$

$$\therefore (1) \Rightarrow f(x) = \frac{1}{2} + \sum_{n=1}^{\infty} \frac{(-1)^n \cos nx}{n^2+1} - \sum_{n=1}^{\infty} (-1)^n \frac{n \sin nx}{n^2+1},$$

$$\text{বা } f(x) = \frac{1}{2} + \left(\frac{-\cos x}{1^2+1} + \frac{\cos 2x}{2^2+1} - \frac{\cos 3x}{3^2+1} + \ldots\right)$$
$$- \left(\frac{-\sin x}{1^2+1} + \frac{2\sin 2x}{2^2+1} - \frac{3\sin 3x}{3^2+1} + \ldots\right).$$

$$\text{বা } f(x) = \frac{1}{2} - \left(\frac{\cos x}{1^2+1} - \frac{\cos 2x}{2^2+1} + \frac{\cos 3x}{3^2+1} - \ldots\right)$$
$$+ \left(\frac{\sin x}{1^2+1} - \frac{2\sin 2x}{2^2+1} + \frac{3\sin 3x}{3^2+1} - \ldots\right).$$

দ্বিতীয় অংশ ঃ [When $x = \pm\pi$ then the sum of the series $= \frac{1}{2}\{f(-\pi+0) + f(\pi-0)\}$

যখন $x = \pm\pi$ তখন ধারাটির যোগফল $= \frac{1}{2}\{f(-\pi+0) + f(\pi-0)\}$.

$$\text{বা ধারাটির যোগফল} = \frac{1}{2}\left(\frac{\pi e^{-\pi}}{2\sinh\pi} + \frac{\pi e^{\pi}}{2\sinh\pi}\right)$$
$$= \frac{\pi}{2\sinh\pi}\left(\frac{e^{\pi}+e^{-\pi}}{2}\right).$$
$$= \frac{\pi\cosh\pi}{2\sinh\pi} = \frac{\pi}{2}\coth\pi.$$

(v). মনে করি [Let] $f(x) = \frac{x(\pi^2-x^2)}{12}$, ... (1)

$$\therefore f(-x) = \frac{-x\{\pi^2-(-x)^2\}}{12} = \frac{-x(\pi^2-x^2)}{12} = -f(x),$$

$\Rightarrow f(x)$ অযুগ্ম ফাংশন, কাজেই ইহার ফ্যুরিয়ার ধারা হইল, [$\Rightarrow f(x)$ is an odd function, so its Fourier series is]

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx, \ldots (2)$$

$$\text{যখন } b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\, dx,$$

$$\text{বা } b_n = \frac{2}{\pi}\int_0^{\pi} \frac{x(\pi^2-x^2)}{12}\sin nx\, dx; \text{ (1) নং দ্বারা।}$$
$$= \frac{1}{6\pi}\int_0^{\pi} (\pi^2 x - x^3)\sin nx\, dx,$$
$$= \frac{1}{6\pi}\left[(\pi^2 x - x^3)\frac{(-\cos nx)}{n} - (\pi^2 - 3x^2)\frac{(-\sin nx)}{n^2} + (-6x)\frac{\cos nx}{n^3} - (-6)\frac{\sin nx}{n^4}\right]_0^{\pi}.$$
$$= \frac{1}{6\pi}\left[\frac{-(\pi^2 x - x^3)\cos nx}{n} + \frac{(\pi^2-3x^2)\sin nx}{n^2} - \frac{6x\cos nx}{n^3} + \frac{6\sin nx}{n^4}\right]_0^{\pi}.$$
$$= \frac{1}{6\pi}\left\{\left(0 + 0 - \frac{6\pi\cos n\pi}{n^3} + 0\right) + 0 - 0 + 0\right\}.$$
$$= -\frac{\cos n\pi}{n^3} = \frac{-(-1)^n}{n^3}.$$

$$\therefore (2) \Rightarrow f(x) = -\sum_{n=1}^{\infty} \frac{(-1)^n \sin nx}{n^3}$$

$$\text{বা } f(x) = -\left(-\frac{\sin x}{1^3} + \frac{\sin 2x}{2^3} - \frac{\sin 3x}{3^3} + \ldots\right)$$

$$\therefore \frac{x(\pi^2-x^2)}{12} = \frac{\sin x}{1^3} - \frac{\sin 2x}{2^3} + \frac{\sin 3x}{3^3} - \ldots, \text{ (1) নং দ্বারা।}$$

(vi). মনে করি [Let] $f(x) = x$, ... (1)

$\therefore f(-x) = -x = -f(x)$.

$\Rightarrow f(x)$ অযুগ্ম ফাংশন, কাজেই ইহার ফ্যুরিয়ার ধারা হইল, [$\Rightarrow f(x)$ is an odd function, so its Fourier series is]

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx, \ldots (2)$$

$$\text{যখন } b_n = \frac{2}{\pi}\int_0^{\pi} f(x)\sin nx\, dx,$$

[illegible] $-\pi < x < \pi$

$x = 2\left(\frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} \ldots\right)$

[illegible] integrate [illegible]

$x^2 = \frac{\pi^2}{3} - 4\left(\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} + \ldots\right)$

[illegible] integrate [illegible]

$\frac{x(\pi^2 - x^2)}{[illegible]} = \frac{\sin x}{1^3} - \frac{\sin 2x}{2^3} + \frac{\sin 3x}{3^3}$

বা $b_n = \frac{2}{\pi}\int_0^{\pi} x \sin nx\, dx$; (1) নং দ্বারা।

$$= \frac{2}{\pi}\left[\frac{x(-\cos nx)}{n} - \frac{(1).(-\sin nx)}{n^2}\right]_0^{\pi}.$$

$$= \frac{2}{\pi}\left[-\frac{x\cos nx}{n} + \frac{\sin nx}{n^2}\right]_0^{\pi}.$$

$$= \frac{2}{\pi}\left\{\left(\frac{-\pi\cos n\pi}{n} + 0\right) - (0)\right\} = \frac{-2\cos n\pi}{n}$$

$$= \frac{-2(-1)^n}{n}.$$

$$\therefore (2) \Rightarrow f(x) = -2\sum_{n=1}^{\infty}\frac{(-1)^n \sin nx}{n}$$

বা $x = -2\left(-\frac{\sin x}{1} + \frac{\sin 2x}{2} - \frac{\sin 3x}{3} + \ldots\right)$; (1) নং দ্বারা।

$$x = 2\left(\frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \ldots\right), \ldots (3)$$

এখন (3) নং এর উভয় পক্ষকে 0 এবং x সীমার মধ্যে x এর সম্পর্কে ইনটিগ্রেট করিয়া পাই, [Now integrating both sides of (3) w. r. to x between the limis 0 and x then we get]

$$\int_0^x x\, dx = 2\int_0^x\left(\frac{\sin x}{1} - \frac{\sin 2x}{2} + \frac{\sin 3x}{3} - \ldots\right) dx.$$

বা $\left[\frac{x^2}{2}\right]_0^x = 2\left[-\frac{\cos x}{1^2} + \frac{\cos 2x}{2^2} - \frac{\cos 3x}{3^2} + \ldots\right]_0^x.$

বা $\left[\frac{x^2}{2}\right]_0^x = -2\left[\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \ldots\right]_0^x.$

বা $\frac{x^2}{2} - 0 = -2\left(\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \ldots\right)$

$$+ 2\left(\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \ldots\right).$$

বা $\frac{x^2}{2} = -2\left(\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \ldots\right) + 2.\frac{\pi^2}{12}.$

বা $x^2 = -4\left(\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \ldots\right) + \frac{\pi^2}{3}, \ldots (4)$

তৃতীয় অংশ ঃ আবার (4) এর উভয় পক্ষেকে 0 এবং x সীমার মধ্যে x এর সম্পর্কে ইনটিগ্রেট করিয়া পাই, [Again integrating both sides of (4) w. r. to x between the limis 0 and x then we get]

$$\int_0^x x^2\, dx = \int_0^x \frac{\pi^2}{3}\, dx - 4\int_0^x\left(\frac{\cos x}{1^2} - \frac{\cos 2x}{2^2} + \frac{\cos 3x}{3^2} - \ldots\right) dx.$$

বা $\left[\frac{x^3}{3}\right]_0^x = \frac{\pi^2}{3}\left[x\right]_0^x - 4\left[\frac{\sin x}{1^3} - \frac{\sin 2x}{2^3} + \frac{\sin 3x}{3^3} - \ldots\right]_0^x.$

বা $\frac{x^3}{3} - \frac{\pi^2 x}{3} = -4\left(\frac{\sin x}{1^3} - \frac{\sin 2x}{2^3} + \frac{\sin 3x}{3^3} - \ldots\right) + 4(0).$

বা $-\frac{x}{3}(\pi^2 - x^2) = -4\left(\frac{\sin x}{1^3} - \frac{\sin 2x}{2^3} + \frac{\sin 3x}{3^3} - \ldots\right).$

বা $\frac{x(\pi^2 - x^2)}{12} = \frac{\sin x}{1^3} - \frac{\sin 2x}{2^3} + \frac{\sin 3x}{3^3} - \ldots$

৪(i). দেওয়া আছে (Given that)

$$f(x) = \begin{cases} l/4, \text{ যখন } -l \le x \le -l/2 \\ x^2/l, \text{ যখন } -l/2 \le x \le l/2 \\ l/4, \text{ যখন } l/2 \le x \le l. \end{cases}$$

যেহেতু $-l \le x \le l$ ব্যবধিতে x এর যোগবোধক এবং বিয়োগবোধক মানের জন্য f(x) এর মান এক রকম, কাজেই f(x) যুগ্ম ফাংশন। সুতরাং ইহার ফ্যুরিয়ার ধারা হইল, [Since the function f(x) has the same value for positive and negative values of x in $-l \le x \le l$, so f(x) is even. Hence the Fourier series is]

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\frac{n\pi x}{l}, \ldots (1)$$

যখন $a_0 = \frac{2}{l}\int_0^l f(x)\, dx$ এবং $a_n = \frac{2}{l}\int_0^l f(x)\cos\frac{n\pi x}{l}\, dx.$